CAD-SEL: Shortcut-Learning Audit and Performance-Campaign Experiment Report

CADSEL Project Workspace

Compiled 2026-09-09 (source: `FINDINGS.md`, last updated 2026-08-30)

Abstract

This report consolidates the **classification** lines of work run in the CAD-SEL project workspace: (1) a ResNet-18 shortcut-learning audit of the dataset, (2) two rounds of classifier-level mitigations (an augmentation-based round and an architectural round with an adversarial discriminator, bbox blurring, and channel attention), (3) a single-modality classification-accuracy campaign (Track A) across four backbone families, and (4) an end-to-end trainable two-tower WLE+EUS fusion architecture. Object detection (the project's Faster R-CNN reproduction grid, its detector-level performance-lever campaign, the detector-ROI multimodal fusion pipeline, and the YOLO11 detection campaign) is out of scope for this report and has been removed. For each remaining line of work we report the model architecture, the augmentation and training hyperparameters, and the quantitative outcome, with an explicit confirmed / confirmed-negative / not-confirmed verdict wherever the underlying work applied a multi-seed noise-floor check. Numbers are drawn from `FINDINGS.md`, cross-checked where needed against the underlying JSON results and source code. This is a curated synthesis organized by line of work, not a line-for-line transcription of every logged run; multi-seed grids are collapsed to mean ± std with the governing verdict.

Contents

1 Introduction **3**
- 1.1 Dataset . . . 3
- 1.2 Reporting metric . . . 3
- 1.3 Discipline applied across the campaign . . . 3
- 1.4 Report organization . . . 3

2 Shortcut-Learning Audit (Experiments 1–6) **4**
- 2.1 Counterfactual ablation protocol . . . 4
- 2.2 Experiments and mechanisms tested . . . 4
- 2.3 Core result (Experiment 3) . . . 4
- 2.4 Mechanism summary . . . 5

3 Mitigation Study I — Augmentation-Level (Experiment 7) **6**
- 3.1 Methods . . . 6
- 3.2 Classifier-level results (Experiment 7) . . . 6

4 Mitigation Study II — Architectural (Experiments 12–16) **7**
- 4.1 Adversarial discriminator with gradient reversal (Experiment 12) . . . 7
- 4.2 Bbox blur (Experiment 13) and channel attention (Experiment 14) . . . 7
- 4.3 Results and verdicts . . . 7

5 Track A — Classification-Accuracy Campaign (§10, §12–§14) **9**
- 5.1 Backbones and infrastructure . . . 9
- 5.2 Fair-baseline gate (10.1) — PASS . . . 9
- 5.3 Backbone screen (10.2) . . . 9
- 5.4 Lever screen and multi-seed confirmation (10.3–10.5) . . . 9

5.5 DINOv3 `-freeze partial` (12.2) — new best classifier . . . 10
5.6 Compounding levers and ensembling (13.2, 13.4) . . . 10
5.7 EUS: treat as shortcut-suspect by default . . . 10
5.8 Closed threads: TTA and DINOv3 fine-tuning LR . . . 10

6 Joint End-to-End Two-Tower WLE+EUS Fusion (§15) 11
6.1 Architecture . . . 11
6.2 Hyperparameters . . . 11
6.3 Binary task: confirmed fusion win . . . 11
6.4 Counterfactual gate — PASSED . . . 12
6.5 Subtype task: confirmed on the unpaired denominator . . . 12
6.6 Negative and null screens . . . 12

7 Cross-Cutting Summary 13
7.1 Campaign-best numbers . . . 13
7.2 Confirmed levers across the whole project . . . 13
7.3 Standing cautions . . . 13

8 Implementation Lessons 14

9 Open Items 15

References 16

A Full Metrics by Run 17
A.1 Track A classifiers (Section 5) — 62 runs . . . 17
A.2 Joint two-tower fusion (Section 6) — 69 run/ablation rows . . . 19
A.3 Shortcut-learning-audit classifier probes (Section 2, 4) — 790 run/variant rows . . . 22

1 Introduction

1.1 Dataset

CAD-SEL is a dual-modal (white-light endoscopy, WLE, and endoscopic ultrasound, EUS) colonoscopy dataset of subepithelial lesions (SELs), collected across 6 acquisition centers.

Property	Value
Patients	641 (537 in Center-1; EUS available for 421, zero EUS-only)
Images	4,912 total, across WLE and EUS
Centers	6 (Center-1 dominates: 81–94% of the training pool depending on split)
Classification subtypes	5 (paper): NET_G1, NET_G2, Leiomyoma, Lipoma, NonNeoplasm
Tasks	`binary` (NET vs. non-NET), `subtype` (5-way patient-level classification)
Class imbalance	∼15:1, NET-heavy (NET_G1 alone is 299 of 641 patients; Leiomyoma is the rarest subtype)
Splitting	patient-level (`patient_key` excludes center, deliberately) unless stated otherwise

1.2 Reporting metric

Every task in this report is scored as **balanced accuracy** (mean per-class recall), because plain accuracy is flattered by always predicting the majority class under ∼15:1 imbalance; macro-F1 is used as a secondary cross-check, and Appendix A additionally reports accuracy, precision, recall, ROC-AUC, and PR-AUC wherever the underlying run logged them.

1.3 Discipline applied across the campaign

Every line of work in this report follows the same protocol, established once a direct measurement showed a single run's own seed-to-seed spread could be larger than several previously reported deltas:

1. A candidate lever is **screened** at a single seed.
2. If directionally positive, it is **confirmed** at ≥3 seeds; a delta only counts if it clears roughly 1–2× the pooled seed-to-seed noise floor, $\sqrt{\sigma_a^2 + \sigma_b^2}$.
3. Any classification-accuracy win is passed through a **mandatory counterfactual gate**: the trained model is re-scored on lesion-removed variants (Section 2) to check the gain is not simply increased shortcut reliance.

1.4 Report organization

Section 2 covers the shortcut-learning audit (Experiments 1–6). Sections 3–4 cover classifier-level mitigations (Experiments 7, 12–16). Section 5 covers the Track A single-modality classification-accuracy campaign. Section 6 covers the end-to-end trainable two-tower WLE+EUS fusion architecture. Section 7 consolidates every confirmed/negative result across these lines of work into master tables.

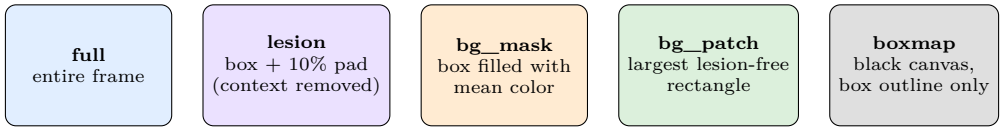
2 Shortcut-Learning Audit (Experiments 1–6)

Full accuracy/balanced-accuracy for every probe and variant in this section: Appendix A.3.

Diagnostics testing whether the label is predictable *without* reading lesion morphology — the failure mode defined by Geirhos et al. (2020). All tests use a ResNet-18 classifier, balanced accuracy, and patient-level splits (unless the experiment is explicitly about splitting).

2.1 Counterfactual ablation protocol

The evidentiary backbone of the whole shortcut audit (and every counterfactual gate used later) is a 5-way image variant, built once per sample and reused verbatim across Experiments 3, 17, 27, and 36:



`bg_mask` is *not* a clean ablation — the filled rectangle still encodes the box's size and position, which Experiment 1 shows is class-associated. `bg_patch` and `boxmap` were added to separate tissue-context leakage from pure geometry leakage; `bg_mask` is retained only to show the contamination it causes.

2.2 Experiments and mechanisms tested

Experiment	Question	Verdict
01 Dataset confounds	Is the label predictable from center, demographics, size, location, geometry?	Center: not present (Cramér's V 0.099)
02 Low-level probes	Is center or the label decodable from global color/exposure stats?	Center: present, strong (0.81 vs. 0.167 chance)
03 Lesion vs. background	Counterfactual ablation: remove the lesion, see what survives	Present, strong (68–95% retained)
04 Cross-center	Train Center-1, test Centers 2–6	Partial (52–100% retained, small n)
05 Split leakage	Patient-level vs. image-level splitting	Present (+0.14 to +0.58 balanced acc.)
06 Synthesis	Collates 01–05 into one verdict table	—

2.3 Core result (Experiment 3)

Balanced accuracy, retention against the full-image reference in brackets:

Task	chance	full	lesion	bg_patch	boxmap
WLE binary	0.500	0.706	0.754 (123%)	**0.695 (95%)**	0.598 (48%)
WLE subtype	0.200	0.326	0.448 (197%)	**0.299 (79%)**	0.300 (80%)
EUS binary	0.500	0.679	0.749 (139%)	**0.621 (68%)**	0.666 (93%)
EUS subtype	0.200	0.322	0.599 (326%)	**0.292 (75%)**	0.431 (189%)

2.4 Mechanism summary

Mechanism	Verdict	Evidence
Center predicts the label	Not present	Cramér's V 0.099; probe lift +0.003
Site encoded in pixels	Present (strong)	center decodable at 0.81 vs. 0.167 chance
Label leaks via global appearance	Not present	max probe lift +0.10
Label predictable from non-lesion pixels	Present (strong)	68–95% retention
Annotation geometry alone diagnostic	Present (strong)	48–189% retention
Collapse on unseen site	Partial	52–100% retained, small external n
Image-level split inflation	Present	+0.14 to +0.58 balanced accuracy

Headline: the label is recoverable from context and box geometry, largely independent of lesion appearance; but the site fingerprint, while real, is *not* the dominant route to the label (Center-1 supplies 537/641 patients, so site and class are near-independent). **Detection is markedly less shortcut-prone than classification**: a classifier retains 68–95% of accuracy on lesion-free crops, while a detector fires on the same crops only $\sim$0.3$\times$ as often as on full frames — box regression forces commitment to a location.

3 Mitigation Study I — Augmentation-Level (Experiment 7)

Full accuracy/balanced-accuracy for every condition/variant here: Appendix A.3.

3.1 Methods

Augmentation	Mechanism
`scale`	Random rescale of the lesion, 0.5–1.6×; targets the geometry shortcut
`bgswap`	Lesion pasted onto a *different patient's* lesion-free background; targets the context shortcut
`both`	`scale` + `bgswap` composed
`hard negatives`	Lesion-free frames from the same patients mined into training, at a target fraction (e.g. 0.3)

Protocol differs from Experiment 3 on purpose: **one** model is trained on ordinary frames and ablated *at inference* ("does this trained model rely on the lesion?"), rather than retraining a separate model per input variant ("is the label learnable from this input?").

3.2 Classifier-level results (Experiment 7)

Change in background reliance (`ret_bg_patch`) vs. baseline, negative = less shortcut:

Task	`scale`	`bgswap`	`both`
WLE binary	−0.6%	−75.2%	−50.9%
WLE subtype	−46.5%	−41.2%	−42.4%
EUS binary	−28.7%	−30.9%	+1.0%
EUS subtype	+37.3%	+26.8%	+23.2%

WLE responds; EUS does not. `both` cuts WLE-binary background reliance from 69% to 18% for −0.069 balanced accuracy; on subtype it cuts 45% to 3% while *improving* accuracy (+0.042). `bgswap` alone is the strongest single intervention (−75.2%) but costs the most accuracy (−0.153). On EUS, every condition makes reliance *worse* (grayscale, narrow-field ultrasound makes a pasted lesion implausible, so the augmentation mostly injects noise).

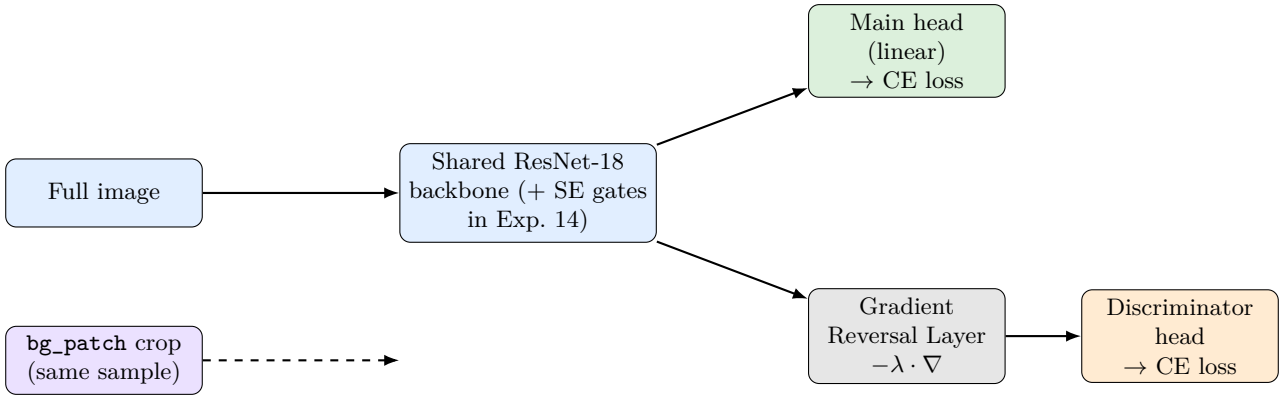
4 Mitigation Study II — Architectural (Experiments 12–16)

Full accuracy/balanced-accuracy for every condition/seed/variant here: Appendix A.3.

Three further classifier-level mitigations, all following Experiment 7's protocol (train *one* model, ablate at inference), reusing Experiment 3's variant definitions and this project's general seed-noise-floor discipline (see the Introduction).

4.1 Adversarial discriminator with gradient reversal (Experiment 12)

A shared ResNet-18 backbone feeds a main classification head (full image, ordinary cross-entropy) and a discriminator head, behind a Gradient Reversal Layer (GRL), that tries to predict the true label from that sample's `bg_patch` crop alone. Reversed gradients push the backbone toward representations from which the discriminator *cannot* predict the label from shortcut-only pixels.



`boxmap` is a secondary, ablatable discriminator target (riskier: a synthetic near-black canvas is out-of-distribution for a natural-image backbone). GRL λ follows a DANN-style sigmoid ramp ($\lambda_{\max} = 1.0$, 2-epoch warm-up).

4.2 Bbox blur (Experiment 13) and channel attention (Experiment 14)

Bbox blur: Gaussian-blurs everything outside a heavily feathered, generously padded ($\geq$1.15× box extent), per-sample-jittered region around the lesion box, *during training only*, guarding explicitly against the `bg_mask` trap (a hard blur boundary would itself encode geometry).

Channel attention: squeeze-and-excite (SE) gates on ResNet-18's stages (simplified from `Channel_Att_Bridge` in UltraLight-VM-UNet). `se_*_ce` (no discriminator) is an *expected-null control*; `se_all_adv` composes SE gates with the Experiment-12 discriminator.

4.3 Results and verdicts

Method	`ret_bg_patch` vs. baseline, 4 modality/task cells	Verdict
Exp. 12 (GRL disc.)	7/8 conditions negative; 1/8 (EUS/subtype) cleared the noise floor	**Not confirmed** at 3 seeds
Exp. 13 (bbox blur)	mixed sign on WLE; **backfires on EUS binary** (+34 to +38% retention)	**No effect**; no boundary-leak fired
Exp. 14 (`se_all_adv`)	**8/8 cells negative**, clean dose-response vs. `se_layer4_adv`	**Not confirmed**, most consistent direction

Exp. 15/16 follow-up. `baseline` and `se_all_adv` extended from 3 to 7 seeds (Exp. 15), then rerun at up to 100 epochs with early-stop patience 20 (Exp. 16). Models converged well before the cap (mean 27–45 epochs). **All 4 cells still negative**, none clearing 2× the pooled seed-noise floor even after proper convergence — noise floors actually *tightened* in 3 of 4 cells. **Final verdict: `se_all_adv` is a real, consistent, but small effect** —

not strong enough to confirm as a validated mitigation. None of the three methods (12/13/14) cleared their own noise floor with the budget tested.

5 Track A — Classification-Accuracy Campaign (§10, §12–§14)

Full accuracy, balanced accuracy, precision, recall, F1, ROC-AUC, and PR-AUC for all 62 runs: Appendix A.1.

Every classification result before this section was a shortcut-learning diagnostic or a frozen-embedding fusion component. Track A pushes single-modality lesion classification (binary NET/non-NET, 5-way subtype) as an end in itself, with a trainable backbone, on the same discipline as the rest of the project.

5.1 Backbones and infrastructure

Backbones	`resnet50`, `convnext_tiny`, `efficientnet_b0`, `dinov3_vits16`
Freeze modes	`none` (full fine-tune), `partial` (last blocks), `head` (frozen backbone)
Losses	`ce`, `weighted_ce`, `seesaw_ce`
Oversampling	`none`, `size` (size-aware RFS, threshold 0.2)
Image size	224
Optimizer	AdamW, lr 3×10^{-4} default, batch 32
Epochs	30 default (20 for the backbone screen, early-stop patience 8)
Cross-validation	5-fold, patient-grouped

Every run writes per-fold loss/LR curves, a confusion matrix, and ROC/PR curves; ROC-AUC/PR-AUC are logged alongside balanced accuracy.

5.2 Fair-baseline gate (10.1) — PASS

A fine-tuned ResNet-50 must clearly beat the frozen-embedding SVM baseline used earlier in this project's audit work (`experiments/svm_classifier.py`, an SVM on frozen ResNet-50/DINOv3 embeddings), or the new pipeline has a bug:

Task	Frozen-SVM	Fine-tuned ResNet-50
WLE binary	0.6447	$\mathbf{0.7616 \pm 0.0234}$
WLE subtype	0.2995	$\mathbf{0.4398 \pm 0.0457}$

5.3 Backbone screen (10.2)

32 runs (4 backbones × 2 modalities × 2 tasks × {frozen, unfrozen}), single seed 42:

Modality/task	Backbone	Frozen (head)	Unfrozen (none)
WLE binary	efficientnet_b0	0.6724	**0.7711**
WLE binary	dinov3_vits16	**0.7072**	0.5711
WLE subtype	efficientnet_b0	0.3069	**0.4348**
EUS binary	resnet50	0.5279	**0.6925**
EUS subtype	efficientnet_b0	0.2327	**0.3602**

(chance: 0.500 binary, 0.200 subtype; full 16-row table in `FINDINGS.md` §10.2.)

efficientnet_b0 is the best unfrozen WLE backbone, carried into the lever screen. **DINOv3 is the one backbone where full fine-tuning backfires** — frozen beats unfrozen on WLE binary (0.7072 vs. 0.5711) and unfrozen collapses to chance on EUS. Root cause identified in §12.3: the project's CNN-tuned default LR (3×10^{-4}) is too high for a ViT.

5.4 Lever screen and multi-seed confirmation (10.3–10.5)

`weighted_ce` on subtype: single-seed +0.0221, confirmed at 3 seeds (0.4282±0.0127 → 0.4609±0.0035, Δ+0.0327, clears floor by ∼2.5×). **Counterfactual gate**: `ret_bg_patch` 33% (win) vs. 42% (baseline) — **lower**, so the gain is not shortcut-driven. **Confirmed** as a validated classification-accuracy improvement.

5.5 DINOv3 `-freeze partial` (12.2) — new best classifier

Unfreezing only the last 4 (of 12) transformer blocks:

Task	seed 42	seed 123	seed 7 (mean ± std)
binary	0.7963	0.8136	0.8105 → **0.8068 ± 0.0075**
subtype	0.4741	0.4899	0.4697 → **0.4779 ± 0.0087**

Confirmed on both tasks (binary +0.0357 over `efficientnet_b0`; subtype +0.0170 over `weighted_ce`). **Counterfactual gate: PARTIAL** — `bg_patch` retention is technically higher than the CNN baselines' (54%>46%, 40%>33%, a literal WARN), but retention on the *isolated-lesion* crop is dramatically higher too (87%/83% vs. 55%/55%) — DINOv3's richer representation appears to extract more signal from *both* lesion and context, not to substitute shortcut-reading for lesion-reading. Recorded as the new best result, flagged partial rather than clean.

5.6 Compounding levers and ensembling (13.2, 13.4)

DINOv3-partial + `weighted_ce` on subtype **Confirmed** at 3 seeds (0.4957 ± 0.0033, gate clean: `bg_patch` retention *dropped* 40%→27%). Multi-seed probability ensembling (3 already-trained seeds, zero extra GPU time) reaches new campaign bests:

Config	seed mean ± std	Ensemble	Δ
DINOv3-partial, binary	0.8063 ± 0.0078	**0.8209**	+0.0146
DINOv3-partial + weighted_ce, subtype	0.4988 ± 0.0049	**0.5230**	+0.0242

5.7 EUS: treat as shortcut-suspect by default

Extending the counterfactual gate to the remaining 17 EUS backbone-screen checkpoints found retentions from −43% to **89%** (`efficientnet_b0`/subtype/frozen) — close to the ∼100% "pure shortcut" DINOv3-EUS precedent from the audit. **No EUS classification result in this campaign is both strong and counterfactually clean**; the single new-best EUS classifier that does clear the gate cleanly is DINOv3 `-freeze partial` (§14.3): binary 0.7386 ± 0.0313 (35% `bg_patch` retention), subtype 0.3764 ± 0.0507 (15% retention).

5.8 Closed threads: TTA and DINOv3 fine-tuning LR

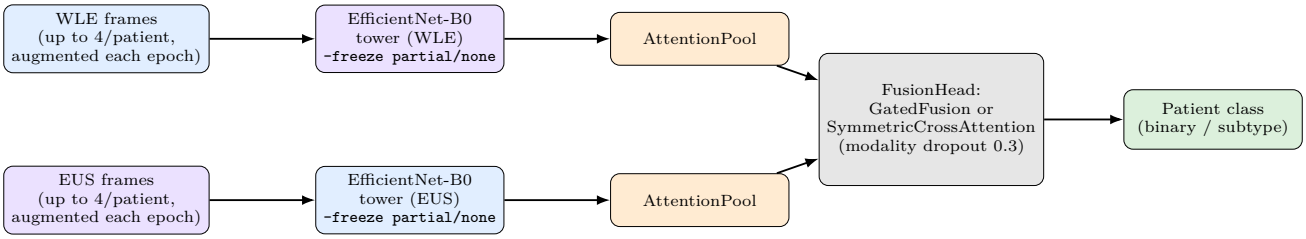
Flip-TTA (12.4) and 5-crop TTA (14.6) are both null on every checkpoint tried: training already includes `RandomHorizontalFlip`/`RandomResizedCrop`, leaving TTA nothing new to add. `unfreeze_last_n` sweep (14.5): $n = 6$–8 modestly beats the default $n = 4$ (single-seed, not confirmed); even $n = 12$ avoids catastrophic collapse because the patch/position embeddings stay frozen under `-freeze partial` regardless of n — the actual reason `-freeze none` was catastrophic (§10.2) while a much lower LR (§12.3, 10^{-5}) recovers most of the gap.

6 Joint End-to-End Two-Tower WLE+EUS Fusion (§15)

Accuracy, balanced accuracy, and macro-F1 for all 69 run/ablation combinations: Appendix A.2.

Every prior fusion result trained `FusionHead` on **frozen, pre-extracted, single-pass** features — the per-modality backbones never see a fusion-loss gradient, and no image augmentation happens in the fusion stage. This architecture is structurally different: two independently trainable backbones, jointly fine-tuned end-to-end, with independent per-modality augmentation every training step.

6.1 Architecture



Two independent `efficientnet_b0` towers (ImageNet-pretrained; `-freeze partial` unfreezes the last two `features` stages) feed the **unmodified** `cadsel.fusion.FusionHead` module: an attention-weighted pool over each modality's frames, then a `GatedFusion` or `SymmetricCrossAttentionFusion` combination layer, with modality dropout and missing-modality masking. One joint model per fold is trained with `modality_dropout=0.3` throughout, and `wle_only` / `eus_only` / `fused` are reported as three masked-inference views of that single model.

6.2 Hyperparameters

Towers	`efficientnet_b0` × 2 (independent weights)
Frame budget	up to 4 frames/patient/modality, sampled with replacement
Loss	`weighted_ce`
Fusion	`gated` (default) or `symmetric`
Learning rate	flat 10^{-4}, or discriminative (backbone 10^{-5} / head 10^{-4})
Epochs	30 (initial screen) or 100 / patience 20 (confirmed configs)
Cross-validation	5-fold, patient-grouped
Optional levers	`-lesion-crop` (crop to box + 10% pad, pre-augmentation), `-bg-remove-prob` (stochastic mean-fill outside the box)

6.3 Binary task: confirmed fusion win

Config	Ablation (denom.)	seed 42/123/7	**3-seed mean ± std**
`-freeze partial`, flat LR	wle_only (unpaired)	0.687/0.722/0.737	0.7153 ± 0.0208
	fused (unpaired)	0.749/0.761/0.769	$\mathbf{0.7596 \pm 0.0082}$
`-freeze none`	wle_only (unpaired)	0.766/0.773/0.754	0.7644 ± 0.0093
	fused (unpaired)	0.803/0.808/0.792	$\mathbf{0.8007 \pm 0.0067}$
`-freeze none` + `-lesion-crop`	wle_only (unpaired)	0.769/0.768/0.783	0.7733 ± 0.0071
	eus_only (paired)	0.682/0.725/0.710	0.7057 ± 0.0178
	fused (unpaired)	0.817/0.794/0.822	$\mathbf{0.8110 \pm 0.0123}$

`fused vs. wle_only` (within `-freeze none`): $\Delta + 0.0363$, noise floor ≈ 0.0115 — **Confirmed**, ∼3.2× the floor, the cleanest margin in this section. **`-freeze none vs. -freeze partial`**, both fused: $\Delta + 0.0411$, floor ≈ 0.0106 — **Confirmed**. `-lesion-crop` confirms a real EUS-only gain ($0.6197 \pm 0.0216 \rightarrow 0.7057 \pm 0.0178$, $\Delta + 0.0860$, ∼3.1× floor) but its `fused` delta over plain `-freeze none` (+0.0103) does *not* clear its own floor (≈ 0.0140) — a real single-modality win that does not (yet) propagate to the headline fused metric, plausibly because `GatedFusion`'s gate structurally favors WLE.

Multi-seed ensembling (zero extra training) reaches campaign-best fused/unpaired balanced accuracy of **0.8219** (`-lesion-crop` ensemble) / 0.8160 (`-freeze none` ensemble alone).

6.4 Counterfactual gate — PASSED

Built `36_joint_fusion_counterfactual.py`, the patient-level, multi-frame, two-modality adaptation of the classifier gate; swaps both towers to the same variant simultaneously per patient. Run against the confirmed `-freeze none` seed-42 checkpoint:

Variant	balanced accuracy	retention vs. full
full (real images, both towers)	0.803	—
lesion (crop to box + padding)	0.632	44%
bg_mask (lesion erased, background kept)	0.644	48%
bg_patch (pure background, no lesion)	0.627	42%
boxmap (black canvas, box outline only)	**0.494**	−2%

With every trace of real tissue removed, balanced accuracy falls to essentially exact chance (0.494 vs. 0.500). **Gate PASSED**: the confirmed win is genuine lesion-dependent signal, not shortcut-reading.

6.5 Subtype task: confirmed on the unpaired denominator

Config	Ablation	seed 42/123/7	**3-seed mean ± std**
discriminative LR, `-freeze partial`	wle_only (unpaired)	0.464/0.445/0.502	0.4704 ± 0.0234
	fused (unpaired)	0.510/0.481/0.528	$\mathbf{0.5063 \pm 0.0193}$
`-freeze none` + `-lesion-crop`	wle_only (unpaired)	0.527/0.469/0.518	0.5049 ± 0.0254
	fused (unpaired)	0.555/0.514/0.557	$\mathbf{0.5419 \pm 0.0201}$

Unpaired balanced accuracy delta +0.0359 vs. floor ≈ 0.0304 (1.18×, tight but real; macro-F1 cross-check confirms more cleanly at 2.9×). `-lesion-crop` on `-freeze none` then confirms a **new subtype best** (0.5419 ± 0.0201, $\Delta + 0.0356$ over the prior best, ∼1.28× floor). Counterfactual gate on the discriminative-LR checkpoint **PASSED** (boxmap retention 2%, essentially chance). **Caveat**: the paired-only denominator's confirmation is razor-thin (clears its floor by 0.0002) and should not be leaned on.

6.6 Negative and null screens

Discriminative backbone LR (§15.3) helped `wle_only` more than `fused`, narrowing rather than widening the fusion gap — a useful negative, not carried forward. `-fusion symmetric` (§15.7) is within noise of gated and does not let lesion-crop's EUS gain reach the fused output either, refuting the specific "WLE-bias" hypothesis. `-bg-remove-prob 0.5` (§15.9) is clearly negative (fused −0.0397), single-seed, not escalated: stochastic mean-filling followed by `RandomResizedCrop` can land entirely inside the filled region, producing noisy supervision rather than useful regularization.

7 Cross-Cutting Summary

7.1 Campaign-best numbers

Line of work	Best confirmed configuration	Score
Track A, WLE binary	DINOv3-partial, 3-seed ensemble	0.8209 bal. acc.
Track A, WLE subtype	DINOv3-partial + weighted_ce, 3-seed ensemble	0.5230 bal. acc.
Track A, EUS binary (gate-clean)	DINOv3-partial	0.7386 ± 0.0313 bal. acc.
Joint fusion, binary	`-freeze none` + `-lesion-crop`, ensemble	0.8219 bal. acc.
Joint fusion, subtype	`-freeze none` + `-lesion-crop`	0.5419 ± 0.0201 bal. acc.

7.2 Confirmed levers across the whole project

Confirmed positive	Confirmed negative
`weighted_ce` loss, WLE subtype (Track A)	DINOv3 `-freeze none` at the project's default LR (Track A)
DINOv3 `-freeze partial` (Track A, both WLE tasks)	Flip-TTA / 5-crop TTA (Track A)
Multi-seed probability ensembling (Track A & joint fusion)	`-bg-remove-prob` (joint fusion)
Joint two-tower fusion, binary (`-freeze none`)	
Joint two-tower fusion, subtype (unpaired denominator)	
`-lesion-crop`, EUS single-modality signal	

7.3 Standing cautions

- **EUS classification results are shortcut-suspect by default** unless individually gated — only DINOv3-partial (§14.3) is both strong and clean.
- **A clean confirmation and a clean counterfactual pass are two different bars.** DINOv3-partial is a confirmed win on both WLE tasks (Section 5) yet only flags PARTIAL on its own counterfactual gate — a result can clear the noise floor without being unambiguously shortcut-free.
- **A confirmation can be real and still be razor-thin.** The joint-fusion subtype win on the paired-only denominator (Section 6) clears its own noise floor by only 0.0002 — treat it as directionally right, not as statistically solid as the unpaired-denominator result next to it.

8 Implementation Lessons

- **Run-name collisions silently overwrite results.** Any CLI flag that can distinguish two otherwise-identical configs must be part of the output filename, or a later run can destroy an earlier one's checkpoints/JSON with no error — seen twice in this project's classification work: a learning-rate sweep silently overwrote Track A's original DINOv3 `-freeze none` failure JSON (§12.3) until `-lr` was added to the run-name template, and the joint-fusion multi-seed ensemble script initially keyed its output only on task, so a second ensemble run overwrote the first's result file (§15.5) until the script was fixed to key on the full ensembled config.

9 Open Items

- A repeated-CV or bootstrapped-confidence-interval protocol for the DINOv3-classifier-backbone fusion rebuild (a frozen-feature variant distinct from the trainable two-tower architecture in Section 6), whose paired-patient pool is too small for a single 5-fold split to give a stable read.
- A counterfactual-gate adaptation for `-lesion-crop`-trained joint-fusion checkpoints (the model never sees uncropped content, so the existing raw-frame gate would confound distribution shift with genuine lesion-dependence).
- Domain-specific backbone pretraining (GastroNet-5M is in-domain for WLE but gated behind a commercial platform; no publicly downloadable SEL dataset currently exists per the paper's own Table 1 comparison).

References

- Xu, L., Chen, S., Li, C. et al. *CAD-SEL: A dual-modal colonoscopy dataset of subepithelial lesion.* Scientific Data (2026). doi:10.1038/s41597-026-07331-y
- Geirhos, R. et al. *Shortcut learning in deep neural networks.* Nature Machine Intelligence (2020).
- Ganin, Y. & Lempitsky, V. *Unsupervised domain adaptation by backpropagation (gradient reversal / DANN).* ICML (2015).
- Wang, J. et al. *Seesaw Loss for Long-Tailed Instance Segmentation.* CVPR (2021).

A Full Metrics by Run

Every table in the main text reports the headline metric(s) actually used to reach each verdict (balanced accuracy), matching how the underlying experiments were designed and gated. This appendix reports **accuracy, balanced accuracy, precision, recall, F1, ROC-AUC, and PR-AUC for every individual classification run** referenced anywhere in this report, extracted programmatically from the project's own saved result files (never re-derived by hand, to avoid transcription error at this scale). Two honesty rules govern every cell below:

- A metric is populated only if the underlying JSON actually contains it, or is exactly derivable from something it contains (precision/recall/F1 from a saved confusion matrix). A cell reading n/l means **not logged** by the original experiment — the script that produced that run never computed or saved that statistic, and no attempt is made to estimate it.

A.1 Track A classifiers (Section 5) — 62 runs

The only line of work in this project with all 7 metrics natively logged per run: accuracy, balanced accuracy, ROC-AUC, and PR-AUC come directly from `experiments/finetune_classifier.py`'s own evaluation code; precision and recall are macro-averaged from each run's saved 5-fold-pooled confusion matrix.

Table 1: Track A classifier runs: full 7-metric breakdown, extracted directly from each run's saved result JSON (`experiments/results/classifier_runs/*.json`). Precision/recall/F1 are macro-averaged over classes, derived from each run's own saved confusion matrix (5-fold pooled). Accuracy and balanced accuracy are the 5-fold mean ± std; ROC-AUC and PR-AUC are the 5-fold mean (per-fold std not separately saved).

Run	Acc.	Bal. Acc.	Prec.	Rec.	F1	ROC-AUC	PR-AUC
convnext_tiny/EUS/binary/freeze-head/os-none/loss-ce/s42	0.792±0.046	0.583±0.057	0.770	0.577	0.579	0.623	0.410
convnext_tiny/EUS/binary/freeze-none/os-none/loss-ce/s42	0.771±0.020	0.692±0.036	0.686	0.700	0.692	0.753	0.508
dinov3_vits16/EUS/binary/freeze-head/os-none/loss-ce/s42	0.778±0.044	0.531±0.039	0.871	0.529	0.492	0.600	0.389
dinov3_vits16/EUS/binary/freeze-none/os-none/loss-ce/s42	0.765±0.038	0.500±0.000	0.382	0.500	0.433	0.538	0.263
dinov3_vits16/EUS/binary/freeze-partial/os-none/loss-ce/s42	0.786±0.024	0.739±0.031	0.711	0.735	0.720	0.782	0.589
efficientnet_b0/EUS/binary/freeze-head/os-none/loss-ce/s42	0.767±0.037	0.576±0.028	0.654	0.580	0.587	0.672	0.397
efficientnet_b0/EUS/binary/freeze-none/os-none/loss-ce/bgswap/s42	0.770±0.027	0.682±0.044	0.680	0.675	0.678	0.753	0.526
efficientnet_b0/EUS/binary/freeze-none/os-none/loss-ce/s42	0.791±0.029	0.684±0.033	0.705	0.681	0.691	0.731	0.543
resnet50/EUS/binary/freeze-head/os-none/loss-ce/s42	0.771±0.031	0.528±0.019	0.698	0.530	0.500	0.601	0.361
resnet50/EUS/binary/freeze-none/os-none/loss-ce/s42	0.782±0.050	0.693±0.021	0.700	0.695	0.697	0.768	0.563
convnext_tiny/EUS/subtype/freeze-head/os-none/loss-ce/s42	0.691±0.061	0.241±0.029	0.262	0.238	0.229	0.616	0.303
convnext_tiny/EUS/subtype/freeze-none/os-none/loss-ce/s42	0.684±0.058	0.330±0.020	0.359	0.322	0.327	0.677	0.363
dinov3_vits16/EUS/subtype/freeze-head/os-none/loss-ce/s42	0.693±0.044	0.205±0.011	0.338	0.205	0.173	0.497	0.238
dinov3_vits16/EUS/subtype/freeze-none/os-none/loss-ce/s42	0.674±0.038	0.233±0.045	0.252	0.227	0.214	0.524	0.247
dinov3_vits16/EUS/subtype/freeze-partial/os-none/loss-ce/s42	0.672±0.034	0.376±0.051	0.405	0.380	0.387	0.719	0.395
efficientnet_b0/EUS/subtype/freeze-head/os-none/loss-ce/s42	0.675±0.041	0.233±0.008	0.288	0.232	0.224	0.609	0.284
efficientnet_b0/EUS/subtype/freeze-none/os-none/loss-ce/bgswap/s42	0.669±0.038	0.371±0.028	0.384	0.372	0.372	0.669	0.375
efficientnet_b0/EUS/subtype/freeze-none/os-none/loss-ce/s42	0.662±0.043	0.360±0.029	0.410	0.366	0.383	0.679	0.392

Run	Acc.	Bal. Acc.	Prec.	Rec.	F1	ROC-AUC	PR-AUC
resnet50/EUS/subtype/freeze-head/os-none/loss-ce/s42	0.690±0.042	0.200±0.000	0.138	0.200	0.163	0.481	0.205
resnet50/EUS/subtype/freeze-none/os-none/loss-ce/s42	0.665±0.049	0.359±0.025	0.417	0.359	0.368	0.652	0.364
convnext_tiny/WLE/binary/freeze-head/os-none/loss-ce/s42	0.773±0.015	0.692±0.029	0.745	0.692	0.707	0.777	0.682
convnext_tiny/WLE/binary/freeze-none/os-none/loss-ce/s42	0.791±0.022	0.761±0.030	0.755	0.760	0.758	0.842	0.735
dinov3_vits16/WLE/binary/freeze-head/os-none/loss-ce/s42	0.795±0.026	0.707±0.044	0.787	0.708	0.727	0.800	0.712
dinov3_vits16/WLE/binary/freeze-none/os-none/loss-ce/lr1e-05/s42	0.814±0.024	0.776±0.027	0.782	0.776	0.779	0.857	0.778
dinov3_vits16/WLE/binary/freeze-partial/os-none/loss-ce/s7	0.834±0.013	0.810±0.015	0.805	0.810	0.807	0.883	0.822
dinov3_vits16/WLE/binary/freeze-partial/os-none/loss-ce/s42	0.822±0.022	0.796±0.019	0.790	0.795	0.793	0.883	0.814
dinov3_vits16/WLE/binary/freeze-partial/os-none/loss-ce/unfreeze12/s42	0.805±0.015	0.767±0.018	0.772	0.767	0.769	0.851	0.769
dinov3_vits16/WLE/binary/freeze-partial/os-none/loss-ce/unfreeze2/s42	0.821±0.019	0.797±0.012	0.789	0.796	0.792	0.878	0.805
dinov3_vits16/WLE/binary/freeze-partial/os-none/loss-ce/unfreeze6/s42	0.826±0.022	0.807±0.013	0.795	0.806	0.800	0.891	0.822
dinov3_vits16/WLE/binary/freeze-partial/os-none/loss-ce/unfreeze8/s42	0.835±0.007	0.808±0.014	0.807	0.808	0.807	0.888	0.825
dinov3_vits16/WLE/binary/freeze-partial/os-none/loss-ce/s123	0.843±0.028	0.814±0.045	0.816	0.813	0.815	0.893	0.828
dinov3_vits16/WLE/binary/freeze-partial/os-none/loss-weighted_ce/s42	0.809±0.030	0.801±0.011	0.777	0.801	0.786	0.880	0.810
efficientnet_b0/WLE/binary/freeze-head/os-none/loss-ce/s42	0.760±0.029	0.672±0.046	0.725	0.672	0.685	0.753	0.641
efficientnet_b0/WLE/binary/freeze-none/os-none/loss-ce/bgswap/s42	0.775±0.017	0.740±0.021	0.737	0.740	0.739	0.814	0.710
efficientnet_b0/WLE/binary/freeze-none/os-none/loss-ce/s42	0.800±0.028	0.771±0.042	0.766	0.771	0.768	0.837	0.750
efficientnet_b0/WLE/binary/freeze-none/os-size/loss-ce/s42	0.809±0.013	0.764±0.027	0.779	0.764	0.771	0.841	0.762
efficientnet_b0/WLE/binary/freeze-none/os-none/loss-seesaw_ce/s42	0.808±0.028	0.760±0.038	0.778	0.761	0.768	0.836	0.762
efficientnet_b0/WLE/binary/freeze-none/os-none/loss-weighted_ce/s42	0.787±0.016	0.764±0.021	0.752	0.764	0.757	0.841	0.757
resnet50/WLE/binary/freeze-head/os-none/loss-ce/s42	0.773±0.015	0.667±0.042	0.768	0.667	0.684	0.739	0.648
resnet50/WLE/binary/freeze-none/os-none/loss-ce/s42	0.795±0.031	0.767±0.028	0.760	0.766	0.763	0.836	0.732
convnext_tiny/WLE/subtype/freeze-head/os-none/loss-ce/s42	0.693±0.016	0.322±0.025	0.451	0.322	0.318	0.700	0.384
convnext_tiny/WLE/subtype/freeze-none/os-none/loss-ce/s42	0.637±0.038	0.433±0.026	0.422	0.430	0.422	0.763	0.446
dinov3_vits16/WLE/subtype/freeze-head/os-none/loss-ce/s42	0.700±0.021	0.311±0.026	0.502	0.311	0.298	0.718	0.406
dinov3_vits16/WLE/subtype/freeze-none/os-none/loss-ce/lr1e-05/s42	0.703±0.009	0.439±0.011	0.464	0.436	0.440	0.763	0.474
dinov3_vits16/WLE/subtype/freeze-partial/os-none/loss-ce/s7	0.707±0.042	0.470±0.031	0.472	0.474	0.469	0.785	0.500
dinov3_vits16/WLE/subtype/freeze-partial/os-none/loss-ce/s42	0.689±0.034	0.474±0.033	0.487	0.474	0.479	0.788	0.504
dinov3_vits16/WLE/subtype/freeze-partial/os-none/loss-ce/s123	0.699±0.031	0.490±0.054	0.498	0.491	0.491	0.782	0.491

Run	Acc.	Bal. Acc.	Prec.	Rec.	F1	ROC-AUC	PR-AUC
dinov3_vits16/WLE/subtype/freeze-partial/os-none/loss-weighted_ce/s7	0.528±0.045	0.494±0.030	0.402	0.492	0.418	0.770	0.471
dinov3_vits16/WLE/subtype/freeze-partial/os-none/loss-weighted_ce/s42	0.574±0.089	0.500±0.025	0.445	0.503	0.460	0.779	0.485
dinov3_vits16/WLE/subtype/freeze-partial/os-none/loss-weighted_ce/s123	0.585±0.095	0.493±0.036	0.428	0.502	0.449	0.782	0.470
efficientnet_b0/WLE/subtype/freeze-head/os-none/loss-ce/s42	0.678±0.013	0.307±0.026	0.398	0.306	0.299	0.683	0.364
efficientnet_b0/WLE/subtype/freeze-none/os-none/loss-ce/s7	0.675±0.025	0.410±0.022	0.448	0.406	0.418	0.728	0.434
efficientnet_b0/WLE/subtype/freeze-none/os-none/loss-ce/bgswap/s42	0.641±0.038	0.432±0.030	0.437	0.430	0.431	0.746	0.449
efficientnet_b0/WLE/subtype/freeze-none/os-none/loss-ce/s42	0.677±0.016	0.435±0.050	0.447	0.433	0.431	0.743	0.466
efficientnet_b0/WLE/subtype/freeze-none/os-none/loss-ce/s123	0.673±0.030	0.439±0.030	0.469	0.440	0.448	0.747	0.454
efficientnet_b0/WLE/subtype/freeze-none/os-size/loss-ce/s42	0.668±0.013	0.416±0.037	0.438	0.413	0.417	0.748	0.437
efficientnet_b0/WLE/subtype/freeze-none/os-none/loss-seesaw_ce/s42	0.691±0.016	0.413±0.041	0.452	0.415	0.419	0.750	0.454
efficientnet_b0/WLE/subtype/freeze-none/os-none/loss-weighted_ce/s7	0.581±0.067	0.466±0.033	0.417	0.466	0.435	0.754	0.450
efficientnet_b0/WLE/subtype/freeze-none/os-none/loss-weighted_ce/s42	0.543±0.043	0.457±0.046	0.390	0.459	0.407	0.735	0.432
efficientnet_b0/WLE/subtype/freeze-none/os-none/loss-weighted_ce/s123	0.578±0.075	0.460±0.038	0.400	0.461	0.418	0.753	0.441
resnet50/WLE/subtype/freeze-head/os-none/loss-ce/s42	0.681±0.024	0.293±0.028	0.400	0.292	0.279	0.687	0.363
resnet50/WLE/subtype/freeze-none/os-none/loss-ce/s42	0.658±0.042	0.416±0.016	0.429	0.415	0.419	0.743	0.439

A.2 Joint two-tower fusion (Section 6) — 69 run/ablation rows

Table 2: Joint end-to-end two-tower fusion runs (§6): accuracy, balanced accuracy, and macro-F1 as logged in `experiments/results/joint_fusion_runs/*.json` (5-fold mean ± std). Precision, recall, ROC-AUC, and PR-AUC were not logged for this architecture (no confusion matrix or class-probability curve was saved per fold) — reported as n/l rather than fabricated.

Run	Ablation	Accuracy	Bal. Acc.	F1 (macro) (Prec./Rec./ROC-AUC/PR-AUC: n/l)
binary__freeze-none__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__bgremove0.3__s42	eus_only	0.690±0.123	0.566±0.084	0.519±0.104
binary__freeze-none__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__bgremove0.3__s42	fused	0.815±0.020	0.779±0.041	0.775±0.028
binary__freeze-none__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__bgremove0.3__s42	wle_only	0.760±0.019	0.757±0.032	0.730±0.021
binary__freeze-none__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__bgremove0.5__s42	eus_only	0.696±0.069	0.640±0.080	0.600±0.073
binary__freeze-none__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__bgremove0.5__s42	fused	0.813±0.023	0.763±0.042	0.766±0.036

Run	Ablation	Accuracy	Bal. Acc.	F1 (macro) (Prec./Rec./ROC-AUC/PR-AUC: n/l)
binary__freeze-none__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__bgremove0.5__s42	wle_only	0.771±0.028	0.736±0.035	0.729±0.032
binary__freeze-none__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__lesioncrop__s123	eus_only	0.744±0.020	0.725±0.036	0.685±0.021
binary__freeze-none__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__lesioncrop__s123	fused	0.821±0.030	0.794±0.012	0.788±0.023
binary__freeze-none__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__lesioncrop__s123	wle_only	0.784±0.053	0.768±0.030	0.750±0.041
binary__freeze-none__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__lesioncrop__s42	eus_only	0.770±0.021	0.682±0.064	0.664±0.050
binary__freeze-none__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__lesioncrop__s42	fused	0.844±0.035	0.817±0.033	0.813±0.039
binary__freeze-none__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__lesioncrop__s42	wle_only	0.776±0.098	0.768±0.061	0.748±0.087
binary__freeze-none__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__lesioncrop__s7	eus_only	0.658±0.056	0.710±0.046	0.625±0.043
binary__freeze-none__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__lesioncrop__s7	fused	0.833±0.022	0.822±0.017	0.806±0.022
binary__freeze-none__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__lesioncrop__s7	wle_only	0.806±0.041	0.783±0.048	0.773±0.047
binary__freeze-none__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__s123	eus_only	0.687±0.049	0.650±0.052	0.617±0.046
binary__freeze-none__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__s123	fused	0.828±0.008	0.808±0.025	0.797±0.014
binary__freeze-none__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__s123	wle_only	0.780±0.006	0.773±0.019	0.750±0.007
binary__freeze-none__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__s42	eus_only	0.707±0.082	0.608±0.030	0.592±0.031
binary__freeze-none__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__s42	fused	0.809±0.028	0.803±0.041	0.780±0.032
binary__freeze-none__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__s42	wle_only	0.753±0.037	0.766±0.043	0.728±0.035
binary__freeze-none__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__s7	eus_only	0.746±0.032	0.601±0.049	0.598±0.047
binary__freeze-none__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__s7	fused	0.839±0.020	0.792±0.015	0.800±0.021
binary__freeze-none__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__s7	wle_only	0.785±0.032	0.754±0.031	0.746±0.032
binary__freeze-none__fusion-symmetric__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__lesioncrop__s42	eus_only	0.710±0.096	0.732±0.062	0.664±0.082
binary__freeze-none__fusion-symmetric__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__lesioncrop__s42	fused	0.818±0.050	0.812±0.026	0.793±0.045
binary__freeze-none__fusion-symmetric__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__lesioncrop__s42	wle_only	0.795±0.025	0.779±0.017	0.763±0.022

Run	Ablation	Accuracy	Bal. Acc.	F1 (macro) (Prec./Rec./ROC-AUC/PR-AUC: n/l)
binary__freeze-partial__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__blr1e-05__s42	eus_only	0.739±0.045	0.585±0.073	0.578±0.066
binary__freeze-partial__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__blr1e-05__s42	fused	0.800±0.044	0.753±0.061	0.753±0.056
binary__freeze-partial__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__blr1e-05__s42	wle_only	0.740±0.037	0.719±0.061	0.701±0.045
binary__freeze-partial__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__paired__s42	eus_only	0.503±0.076	0.628±0.063	0.495±0.071
binary__freeze-partial__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__paired__s42	fused	0.770±0.062	0.763±0.043	0.717±0.051
binary__freeze-partial__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__paired__s42	wle_only	0.782±0.033	0.699±0.057	0.693±0.047
binary__freeze-partial__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__s123	eus_only	0.735±0.061	0.566±0.075	0.525±0.085
binary__freeze-partial__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__s123	fused	0.811±0.024	0.761±0.018	0.765±0.023
binary__freeze-partial__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__s123	wle_only	0.755±0.047	0.722±0.017	0.712±0.035
binary__freeze-partial__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__s42	eus_only	0.716±0.054	0.641±0.113	0.590±0.104
binary__freeze-partial__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__s42	fused	0.779±0.059	0.749±0.073	0.740±0.069
binary__freeze-partial__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__s42	wle_only	0.679±0.128	0.687±0.088	0.647±0.120
binary__freeze-partial__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__s7	eus_only	0.687±0.050	0.610±0.028	0.596±0.034
binary__freeze-partial__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__s7	fused	0.814±0.036	0.769±0.043	0.771±0.044
binary__freeze-partial__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__s7	wle_only	0.767±0.051	0.737±0.051	0.727±0.054
binary__freeze-partial__fusion-symmetric__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__s42	eus_only	0.571±0.120	0.598±0.053	0.507±0.063
binary__freeze-partial__fusion-symmetric__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__s42	fused	0.779±0.029	0.759±0.031	0.743±0.021
binary__freeze-partial__fusion-symmetric__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__s42	wle_only	0.748±0.045	0.714±0.038	0.705±0.041
subtype__freeze-none__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__lesioncrop__s123	eus_only	0.614±0.083	0.371±0.050	0.296±0.036
subtype__freeze-none__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__lesioncrop__s123	fused	0.510±0.085	0.513±0.058	0.401±0.014
subtype__freeze-none__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__lesioncrop__s123	wle_only	0.416±0.083	0.469±0.069	0.347±0.020
subtype__freeze-none__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__lesioncrop__s42	eus_only	0.566±0.177	0.371±0.038	0.286±0.034

Run	Ablation	Accuracy	Bal. Acc.	F1 (macro) (Prec./Rec./ROC-AUC/PR-AUC: n/l)
subtype__freeze-none__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__lesioncrop__s42	fused	0.549±0.044	0.555±0.034	0.453±0.027
subtype__freeze-none__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__lesioncrop__s42	wle_only	0.464±0.057	0.527±0.041	0.408±0.038
subtype__freeze-none__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__lesioncrop__s7	eus_only	0.531±0.126	0.350±0.049	0.260±0.042
subtype__freeze-none__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__lesioncrop__s7	fused	0.551±0.091	0.557±0.035	0.460±0.057
subtype__freeze-none__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__lesioncrop__s7	wle_only	0.465±0.077	0.518±0.035	0.401±0.034
subtype__freeze-partial__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__blr1e-05__s123	eus_only	0.460±0.174	0.232±0.068	0.210±0.039
subtype__freeze-partial__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__blr1e-05__s123	fused	0.508±0.059	0.481±0.051	0.407±0.054
subtype__freeze-partial__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__blr1e-05__s123	wle_only	0.411±0.045	0.445±0.051	0.351±0.047
subtype__freeze-partial__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__blr1e-05__s42	eus_only	0.437±0.144	0.242±0.033	0.189±0.026
subtype__freeze-partial__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__blr1e-05__s42	fused	0.430±0.081	0.510±0.048	0.391±0.042
subtype__freeze-partial__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__blr1e-05__s42	wle_only	0.325±0.045	0.464±0.036	0.310±0.033
subtype__freeze-partial__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__blr1e-05__s7	eus_only	0.493±0.118	0.303±0.082	0.258±0.046
subtype__freeze-partial__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__blr1e-05__s7	fused	0.481±0.039	0.528±0.035	0.418±0.022
subtype__freeze-partial__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__blr1e-05__s7	wle_only	0.381±0.062	0.502±0.015	0.354±0.027
subtype__freeze-partial__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__s42	eus_only	0.450±0.053	0.318±0.077	0.225±0.055
subtype__freeze-partial__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__s42	fused	0.500±0.040	0.496±0.037	0.404±0.029
subtype__freeze-partial__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__ep100__es20__s42	wle_only	0.396±0.051	0.459±0.055	0.346±0.036
subtype__freeze-partial__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__s42	eus_only	0.387±0.061	0.251±0.057	0.185±0.038
subtype__freeze-partial__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__s42	fused	0.492±0.047	0.501±0.015	0.395±0.023
subtype__freeze-partial__fusion-gated__loss-weighted_ce__moddrop0.3__mf4__s42	wle_only	0.398±0.052	0.464±0.040	0.349±0.044

A.3 Shortcut-learning-audit classifier probes (Section 2, 4) — 790 run/variant rows

Experiments 1–5, 7, and 12–14/17–18: ResNet-18 counterfactual probes and classifier-level mitigation screens. These predate Track A's infrastructure and were built around a single metric (balanced accuracy against a chance/majority floor) by design — see `experiments/probes.py`'s own docstring on why this line of work deliberately avoids scikit-learn. No confusion matrix, class-probability score, or checkpoint was retained, so precision/recall/F1/ROC-AUC/PR-AUC cannot be recovered without fully retraining roughly 150 individual

model fits; per the scope agreed for this report, that retraining was not performed.

Table 3: Shortcut-learning-audit and classifier-level mitigation probes (Experiments 1–5, 7, 12–14, 17–18): accuracy and balanced accuracy as logged in each experiment's own `experiments/results/*.json`. These ResNet-18 probe scripts predate Track A's infrastructure and never saved a confusion matrix, class-probability scores, or a checkpoint (they resume from JSON only, per `experiments/README.md`) — precision, recall, F1, ROC-AUC, and PR-AUC were never computed for these runs, and backfilling them would require fully retraining every one of the ∼150 individual model fits below. Reported honestly as n/l rather than estimated.

Exp.	Mod.	Task	Configuration	Acc.	Bal. Acc.	P/R/F1/ROC/PR-AUC
01	–	category	metadata: centre only	0.123	0.203	n/l
01	–	category	metadata: demographics	0.251	0.268	n/l
01	–	category	metadata: lesion size	0.474	0.402	n/l
01	–	category	metadata: location	0.309	0.403	n/l
01	–	category	metadata: all metadata	0.602	0.387	n/l
01	–	is_net	metadata: centre only	0.395	0.496	n/l
01	–	is_net	metadata: lesion size	0.743	0.679	n/l
01	–	is_net	metadata: all metadata	0.844	0.805	n/l
01	WLE	subtype	geometry: None	0.207	0.208	n/l
01	WLE	is_net	geometry: None	0.567	0.541	n/l
01	WLE	center	geometry: None	0.846	0.832	n/l
01	EUS	subtype	geometry: None	0.139	0.268	n/l
01	EUS	is_net	geometry: None	0.705	0.578	n/l
01	EUS	center	geometry: None	0.977	0.897	n/l
02	WLE	center	A/MLP	0.952	0.754	n/l
02	WLE	center	A/RF	0.974	0.809	n/l
02	EUS	center	A/MLP	0.942	0.696	n/l
02	EUS	center	A/RF	0.972	0.782	n/l
02	WLE	subtype	B/MLP	0.478	0.248	n/l
02	WLE	subtype	B/RF	0.646	0.254	n/l
02	WLE	is_net	B/MLP	0.642	0.579	n/l
02	WLE	is_net	B/RF	0.728	0.603	n/l
02	EUS	subtype	B/MLP	0.454	0.228	n/l
02	EUS	subtype	B/RF	0.629	0.198	n/l
02	EUS	is_net	B/MLP	0.630	0.528	n/l
02	EUS	is_net	B/RF	0.723	0.514	n/l
02	WLE	subtype	C/RF	0.643	0.253	n/l
02	WLE	is_net	C/RF	0.716	0.588	n/l
02	WLE	center	C/RF	0.956	0.642	n/l
02	EUS	subtype	C/RF	0.608	0.199	n/l
02	EUS	is_net	C/RF	0.711	0.496	n/l
02	EUS	center	C/RF	0.733	0.447	n/l
03	WLE	binary	full	0.775	0.706	n/l
03	WLE	binary	lesion	0.805	0.754	n/l
03	WLE	binary	bg_mask	0.736	0.667	n/l
03	WLE	binary	bg_patch	0.757	0.695	n/l
03	WLE	binary	boxmap	0.680	0.598	n/l
03	WLE	subtype	full	0.435	0.326	n/l
03	WLE	subtype	lesion	0.664	0.448	n/l
03	WLE	subtype	bg_mask	0.317	0.347	n/l
03	WLE	subtype	bg_patch	0.252	0.299	n/l
03	WLE	subtype	boxmap	0.319	0.300	n/l
03	EUS	binary	full	0.775	0.679	n/l
03	EUS	binary	lesion	0.855	0.749	n/l
03	EUS	binary	bg_mask	0.793	0.766	n/l
03	EUS	binary	bg_patch	0.561	0.621	n/l
03	EUS	binary	boxmap	0.717	0.666	n/l
03	EUS	subtype	full	0.694	0.322	n/l
03	EUS	subtype	lesion	0.395	0.599	n/l
03	EUS	subtype	bg_mask	0.522	0.396	n/l
03	EUS	subtype	bg_patch	0.375	0.292	n/l
03	EUS	subtype	boxmap	0.223	0.431	n/l
04	WLE	binary	internal	0.749	0.685	n/l
04	WLE	binary	external	0.681	0.632	n/l
04	WLE	subtype	internal	0.582	0.395	n/l
04	WLE	subtype	external	0.652	0.300	n/l
04	EUS	binary	internal	0.780	0.636	n/l

Exp.	Mod.	Task	Configuration	Acc.	Bal. Acc.	P/R/F1/ROC/PR-AUC
04	EUS	binary	external	0.745	0.635	n/l
04	EUS	subtype	internal	0.627	0.236	n/l
04	EUS	subtype	external	0.575	0.261	n/l
05	WLE	binary	patient	0.775	0.706	n/l
05	WLE	subtype	patient	0.435	0.326	n/l
05	WLE	binary	image	0.883	0.849	n/l
05	WLE	subtype	image	0.801	0.682	n/l
05	EUS	binary	patient	0.775	0.679	n/l
05	EUS	subtype	patient	0.694	0.322	n/l
05	EUS	binary	image	0.967	0.954	n/l
05	EUS	subtype	image	0.950	0.900	n/l
07	WLE	binary	baseline / full	n/l	0.706	n/l
07	WLE	binary	baseline / bg_patch	n/l	0.642	n/l
07	WLE	binary	baseline / boxmap	n/l	0.500	n/l
07	WLE	binary	scale / full	n/l	0.693	n/l
07	WLE	binary	scale / bg_patch	n/l	0.632	n/l
07	WLE	binary	scale / boxmap	n/l	0.564	n/l
07	WLE	binary	bgswap / full	n/l	0.553	n/l
07	WLE	binary	bgswap / bg_patch	n/l	0.497	n/l
07	WLE	binary	bgswap / boxmap	n/l	0.463	n/l
07	WLE	binary	both / full	n/l	0.637	n/l
07	WLE	binary	both / bg_patch	n/l	0.525	n/l
07	WLE	binary	both / boxmap	n/l	0.500	n/l
07	WLE	subtype	baseline / full	n/l	0.326	n/l
07	WLE	subtype	baseline / bg_patch	n/l	0.257	n/l
07	WLE	subtype	baseline / boxmap	n/l	0.184	n/l
07	WLE	subtype	scale / full	n/l	0.324	n/l
07	WLE	subtype	scale / bg_patch	n/l	0.198	n/l
07	WLE	subtype	scale / boxmap	n/l	0.169	n/l
07	WLE	subtype	bgswap / full	n/l	0.311	n/l
07	WLE	subtype	bgswap / bg_patch	n/l	0.204	n/l
07	WLE	subtype	bgswap / boxmap	n/l	0.302	n/l
07	WLE	subtype	both / full	n/l	0.368	n/l
07	WLE	subtype	both / bg_patch	n/l	0.204	n/l
07	WLE	subtype	both / boxmap	n/l	0.237	n/l
07	EUS	binary	baseline / full	n/l	0.679	n/l
07	EUS	binary	baseline / bg_patch	n/l	0.585	n/l
07	EUS	binary	baseline / boxmap	n/l	0.500	n/l
07	EUS	binary	scale / full	n/l	0.697	n/l
07	EUS	binary	scale / bg_patch	n/l	0.536	n/l
07	EUS	binary	scale / boxmap	n/l	0.412	n/l
07	EUS	binary	bgswap / full	n/l	0.591	n/l
07	EUS	binary	bgswap / bg_patch	n/l	0.515	n/l
07	EUS	binary	bgswap / boxmap	n/l	0.524	n/l
07	EUS	binary	both / full	n/l	0.577	n/l
07	EUS	binary	both / bg_patch	n/l	0.537	n/l
07	EUS	binary	both / boxmap	n/l	0.538	n/l
07	EUS	subtype	baseline / full	n/l	0.322	n/l
07	EUS	subtype	baseline / bg_patch	n/l	0.244	n/l
07	EUS	subtype	baseline / boxmap	n/l	0.330	n/l
07	EUS	subtype	scale / full	n/l	0.278	n/l
07	EUS	subtype	scale / bg_patch	n/l	0.257	n/l
07	EUS	subtype	scale / boxmap	n/l	0.155	n/l
07	EUS	subtype	bgswap / full	n/l	0.304	n/l
07	EUS	subtype	bgswap / bg_patch	n/l	0.266	n/l
07	EUS	subtype	bgswap / boxmap	n/l	0.244	n/l
07	EUS	subtype	both / full	n/l	0.255	n/l
07	EUS	subtype	both / bg_patch	n/l	0.233	n/l
07	EUS	subtype	both / boxmap	n/l	0.216	n/l
12	WLE	subtype	baseline (s42) / full	n/l	0.349	n/l
12	WLE	subtype	baseline (s42) / bg_patch	n/l	0.222	n/l
12	WLE	subtype	baseline (s42) / boxmap	n/l	0.236	n/l
12	WLE	subtype	presweep_l0.3_bg_patch (s42) / full	n/l	0.352	n/l
12	WLE	subtype	presweep_l0.3_bg_patch (s42) / bg_patch	n/l	0.219	n/l
12	WLE	subtype	presweep_l0.3_bg_patch (s42) / boxmap	n/l	0.232	n/l
12	WLE	subtype	presweep_l0.3_bg_patch-boxmap (s42) / full	n/l	0.366	n/l

Exp.	Mod.	Task	Configuration	Acc.	Bal. Acc.	P/R/F1/ROC/PR-AUC
12	WLE	subtype	presweep_l0.3_bg_patch-boxmap (s42) / bg_patch	n/l	0.270	n/l
12	WLE	subtype	presweep_l0.3_bg_patch-boxmap (s42) / boxmap	n/l	0.203	n/l
12	WLE	subtype	presweep_l1.0_bg_patch (s42) / full	n/l	0.352	n/l
12	WLE	subtype	presweep_l1.0_bg_patch (s42) / bg_patch	n/l	0.219	n/l
12	WLE	subtype	presweep_l1.0_bg_patch (s42) / boxmap	n/l	0.232	n/l
12	WLE	subtype	presweep_l1.0_bg_patch-boxmap (s42) / full	n/l	0.366	n/l
12	WLE	subtype	presweep_l1.0_bg_patch-boxmap (s42) / bg_patch	n/l	0.270	n/l
12	WLE	subtype	presweep_l1.0_bg_patch-boxmap (s42) / boxmap	n/l	0.203	n/l
12	WLE	subtype	presweep_l3.0_bg_patch (s42) / full	n/l	0.352	n/l
12	WLE	subtype	presweep_l3.0_bg_patch (s42) / bg_patch	n/l	0.219	n/l
12	WLE	subtype	presweep_l3.0_bg_patch (s42) / boxmap	n/l	0.232	n/l
12	WLE	subtype	presweep_l3.0_bg_patch-boxmap (s42) / full	n/l	0.366	n/l
12	WLE	subtype	presweep_l3.0_bg_patch-boxmap (s42) / bg_patch	n/l	0.270	n/l
12	WLE	subtype	presweep_l3.0_bg_patch-boxmap (s42) / boxmap	n/l	0.203	n/l
12	WLE	binary	baseline (s42) / full	n/l	0.728	n/l
12	WLE	binary	baseline (s42) / bg_patch	n/l	0.653	n/l
12	WLE	binary	baseline (s42) / boxmap	n/l	0.511	n/l
12	WLE	binary	baseline (s7) / full	n/l	0.722	n/l
12	WLE	binary	baseline (s7) / bg_patch	n/l	0.572	n/l
12	WLE	binary	baseline (s7) / boxmap	n/l	0.499	n/l
12	WLE	binary	baseline (s123) / full	n/l	0.705	n/l
12	WLE	binary	baseline (s123) / bg_patch	n/l	0.619	n/l
12	WLE	binary	baseline (s123) / boxmap	n/l	0.509	n/l
12	WLE	binary	adv_bgpatch (s42) / full	n/l	0.692	n/l
12	WLE	binary	adv_bgpatch (s42) / bg_patch	n/l	0.589	n/l
12	WLE	binary	adv_bgpatch (s42) / boxmap	n/l	0.503	n/l
12	WLE	binary	adv_bgpatch (s7) / full	n/l	0.700	n/l
12	WLE	binary	adv_bgpatch (s7) / bg_patch	n/l	0.592	n/l
12	WLE	binary	adv_bgpatch (s7) / boxmap	n/l	0.481	n/l
12	WLE	binary	adv_bgpatch (s123) / full	n/l	0.699	n/l
12	WLE	binary	adv_bgpatch (s123) / bg_patch	n/l	0.567	n/l
12	WLE	binary	adv_bgpatch (s123) / boxmap	n/l	0.500	n/l
12	WLE	binary	adv_bgpatch_boxmap (s42) / full	n/l	0.667	n/l
12	WLE	binary	adv_bgpatch_boxmap (s42) / bg_patch	n/l	0.558	n/l
12	WLE	binary	adv_bgpatch_boxmap (s42) / boxmap	n/l	0.462	n/l
12	WLE	binary	adv_bgpatch_boxmap (s7) / full	n/l	0.660	n/l
12	WLE	binary	adv_bgpatch_boxmap (s7) / bg_patch	n/l	0.580	n/l
12	WLE	binary	adv_bgpatch_boxmap (s7) / boxmap	n/l	0.502	n/l
12	WLE	binary	adv_bgpatch_boxmap (s123) / full	n/l	0.693	n/l
12	WLE	binary	adv_bgpatch_boxmap (s123) / bg_patch	n/l	0.532	n/l
12	WLE	binary	adv_bgpatch_boxmap (s123) / boxmap	n/l	0.500	n/l
12	WLE	subtype	baseline (s7) / full	n/l	0.359	n/l
12	WLE	subtype	baseline (s7) / bg_patch	n/l	0.255	n/l
12	WLE	subtype	baseline (s7) / boxmap	n/l	0.200	n/l
12	WLE	subtype	baseline (s123) / full	n/l	0.327	n/l
12	WLE	subtype	baseline (s123) / bg_patch	n/l	0.300	n/l
12	WLE	subtype	baseline (s123) / boxmap	n/l	0.197	n/l
12	WLE	subtype	adv_bgpatch (s42) / full	n/l	0.352	n/l
12	WLE	subtype	adv_bgpatch (s42) / bg_patch	n/l	0.219	n/l
12	WLE	subtype	adv_bgpatch (s42) / boxmap	n/l	0.232	n/l
12	WLE	subtype	adv_bgpatch (s7) / full	n/l	0.375	n/l
12	WLE	subtype	adv_bgpatch (s7) / bg_patch	n/l	0.262	n/l
12	WLE	subtype	adv_bgpatch (s7) / boxmap	n/l	0.199	n/l
12	WLE	subtype	adv_bgpatch (s123) / full	n/l	0.349	n/l
12	WLE	subtype	adv_bgpatch (s123) / bg_patch	n/l	0.251	n/l
12	WLE	subtype	adv_bgpatch (s123) / boxmap	n/l	0.191	n/l
12	WLE	subtype	adv_bgpatch_boxmap (s42) / full	n/l	0.366	n/l
12	WLE	subtype	adv_bgpatch_boxmap (s42) / bg_patch	n/l	0.270	n/l

Exp.	Mod.	Task	Configuration	Acc.	Bal. Acc.	P/R/F1/ROC/PR-AUC
12	WLE	subtype	adv_bgpatch_boxmap (s42) / boxmap	n/l	0.203	n/l
12	WLE	subtype	adv_bgpatch_boxmap (s7) / full	n/l	0.386	n/l
12	WLE	subtype	adv_bgpatch_boxmap (s7) / bg_patch	n/l	0.260	n/l
12	WLE	subtype	adv_bgpatch_boxmap (s7) / boxmap	n/l	0.200	n/l
12	WLE	subtype	adv_bgpatch_boxmap (s123) / full	n/l	0.362	n/l
12	WLE	subtype	adv_bgpatch_boxmap (s123) / bg_patch	n/l	0.193	n/l
12	WLE	subtype	adv_bgpatch_boxmap (s123) / boxmap	n/l	0.224	n/l
12	EUS	binary	baseline (s42) / full	n/l	0.656	n/l
12	EUS	binary	baseline (s42) / bg_patch	n/l	0.527	n/l
12	EUS	binary	baseline (s42) / boxmap	n/l	0.486	n/l
12	EUS	binary	baseline (s7) / full	n/l	0.694	n/l
12	EUS	binary	baseline (s7) / bg_patch	n/l	0.584	n/l
12	EUS	binary	baseline (s7) / boxmap	n/l	0.500	n/l
12	EUS	binary	baseline (s123) / full	n/l	0.649	n/l
12	EUS	binary	baseline (s123) / bg_patch	n/l	0.592	n/l
12	EUS	binary	baseline (s123) / boxmap	n/l	0.500	n/l
12	EUS	binary	adv_bgpatch (s42) / full	n/l	0.583	n/l
12	EUS	binary	adv_bgpatch (s42) / bg_patch	n/l	0.558	n/l
12	EUS	binary	adv_bgpatch (s42) / boxmap	n/l	0.500	n/l
12	EUS	binary	adv_bgpatch (s7) / full	n/l	0.639	n/l
12	EUS	binary	adv_bgpatch (s7) / bg_patch	n/l	0.549	n/l
12	EUS	binary	adv_bgpatch (s7) / boxmap	n/l	0.500	n/l
12	EUS	binary	adv_bgpatch (s123) / full	n/l	0.620	n/l
12	EUS	binary	adv_bgpatch (s123) / bg_patch	n/l	0.604	n/l
12	EUS	binary	adv_bgpatch (s123) / boxmap	n/l	0.500	n/l
12	EUS	binary	adv_bgpatch_boxmap (s42) / full	n/l	0.567	n/l
12	EUS	binary	adv_bgpatch_boxmap (s42) / bg_patch	n/l	0.526	n/l
12	EUS	binary	adv_bgpatch_boxmap (s42) / boxmap	n/l	0.487	n/l
12	EUS	binary	adv_bgpatch_boxmap (s7) / full	n/l	0.639	n/l
12	EUS	binary	adv_bgpatch_boxmap (s7) / bg_patch	n/l	0.515	n/l
12	EUS	binary	adv_bgpatch_boxmap (s7) / boxmap	n/l	0.499	n/l
12	EUS	binary	adv_bgpatch_boxmap (s123) / full	n/l	0.583	n/l
12	EUS	binary	adv_bgpatch_boxmap (s123) / bg_patch	n/l	0.502	n/l
12	EUS	binary	adv_bgpatch_boxmap (s123) / boxmap	n/l	0.496	n/l
12	EUS	subtype	baseline (s42) / full	n/l	0.374	n/l
12	EUS	subtype	baseline (s42) / bg_patch	n/l	0.318	n/l
12	EUS	subtype	baseline (s42) / boxmap	n/l	0.277	n/l
12	EUS	subtype	baseline (s7) / full	n/l	0.323	n/l
12	EUS	subtype	baseline (s7) / bg_patch	n/l	0.230	n/l
12	EUS	subtype	baseline (s7) / boxmap	n/l	0.200	n/l
12	EUS	subtype	baseline (s123) / full	n/l	0.286	n/l
12	EUS	subtype	baseline (s123) / bg_patch	n/l	0.301	n/l
12	EUS	subtype	baseline (s123) / boxmap	n/l	0.208	n/l
12	EUS	subtype	adv_bgpatch (s42) / full	n/l	0.377	n/l
12	EUS	subtype	adv_bgpatch (s42) / bg_patch	n/l	0.334	n/l
12	EUS	subtype	adv_bgpatch (s42) / boxmap	n/l	0.254	n/l
12	EUS	subtype	adv_bgpatch (s7) / full	n/l	0.282	n/l
12	EUS	subtype	adv_bgpatch (s7) / bg_patch	n/l	0.247	n/l
12	EUS	subtype	adv_bgpatch (s7) / boxmap	n/l	0.217	n/l
12	EUS	subtype	adv_bgpatch (s123) / full	n/l	0.354	n/l
12	EUS	subtype	adv_bgpatch (s123) / bg_patch	n/l	0.228	n/l
12	EUS	subtype	adv_bgpatch (s123) / boxmap	n/l	0.203	n/l
12	EUS	subtype	adv_bgpatch_boxmap (s42) / full	n/l	0.301	n/l
12	EUS	subtype	adv_bgpatch_boxmap (s42) / bg_patch	n/l	0.193	n/l
12	EUS	subtype	adv_bgpatch_boxmap (s42) / boxmap	n/l	0.246	n/l
12	EUS	subtype	adv_bgpatch_boxmap (s7) / full	n/l	0.277	n/l
12	EUS	subtype	adv_bgpatch_boxmap (s7) / bg_patch	n/l	0.186	n/l
12	EUS	subtype	adv_bgpatch_boxmap (s7) / boxmap	n/l	0.200	n/l
12	EUS	subtype	adv_bgpatch_boxmap (s123) / full	n/l	0.318	n/l
12	EUS	subtype	adv_bgpatch_boxmap (s123) / bg_patch	n/l	0.223	n/l
12	EUS	subtype	adv_bgpatch_boxmap (s123) / boxmap	n/l	0.279	n/l
12	WLE	binary	baseline (s1) / full	n/l	0.724	n/l
12	WLE	binary	baseline (s1) / bg_patch	n/l	0.642	n/l
12	WLE	binary	baseline (s1) / boxmap	n/l	0.502	n/l
12	WLE	binary	baseline (s2) / full	n/l	0.726	n/l
12	WLE	binary	baseline (s2) / bg_patch	n/l	0.625	n/l
12	WLE	binary	baseline (s2) / boxmap	n/l	0.483	n/l
12	WLE	binary	baseline (s3) / full	n/l	0.744	n/l

Exp.	Mod.	Task	Configuration	Acc.	Bal. Acc.	P/R/F1/ROC/PR-AUC
12	WLE	binary	baseline (s3) / bg_patch	n/l	0.610	n/l
12	WLE	binary	baseline (s3) / boxmap	n/l	0.475	n/l
12	WLE	binary	baseline (s4) / full	n/l	0.699	n/l
12	WLE	binary	baseline (s4) / bg_patch	n/l	0.611	n/l
12	WLE	binary	baseline (s4) / boxmap	n/l	0.531	n/l
12	WLE	subtype	baseline (s1) / full	n/l	0.389	n/l
12	WLE	subtype	baseline (s1) / bg_patch	n/l	0.202	n/l
12	WLE	subtype	baseline (s1) / boxmap	n/l	0.203	n/l
12	WLE	subtype	baseline (s2) / full	n/l	0.353	n/l
12	WLE	subtype	baseline (s2) / bg_patch	n/l	0.258	n/l
12	WLE	subtype	baseline (s2) / boxmap	n/l	0.204	n/l
12	WLE	subtype	baseline (s3) / full	n/l	0.344	n/l
12	WLE	subtype	baseline (s3) / bg_patch	n/l	0.285	n/l
12	WLE	subtype	baseline (s3) / boxmap	n/l	0.182	n/l
12	WLE	subtype	baseline (s4) / full	n/l	0.369	n/l
12	WLE	subtype	baseline (s4) / bg_patch	n/l	0.278	n/l
12	WLE	subtype	baseline (s4) / boxmap	n/l	0.199	n/l
12	EUS	binary	baseline (s1) / full	n/l	0.666	n/l
12	EUS	binary	baseline (s1) / bg_patch	n/l	0.566	n/l
12	EUS	binary	baseline (s1) / boxmap	n/l	0.526	n/l
12	EUS	binary	baseline (s2) / full	n/l	0.679	n/l
12	EUS	binary	baseline (s2) / bg_patch	n/l	0.576	n/l
12	EUS	binary	baseline (s2) / boxmap	n/l	0.500	n/l
12	EUS	binary	baseline (s3) / full	n/l	0.694	n/l
12	EUS	binary	baseline (s3) / bg_patch	n/l	0.556	n/l
12	EUS	binary	baseline (s3) / boxmap	n/l	0.412	n/l
12	EUS	binary	baseline (s4) / full	n/l	0.695	n/l
12	EUS	binary	baseline (s4) / bg_patch	n/l	0.472	n/l
12	EUS	binary	baseline (s4) / boxmap	n/l	0.444	n/l
12	EUS	subtype	baseline (s1) / full	n/l	0.322	n/l
12	EUS	subtype	baseline (s1) / bg_patch	n/l	0.226	n/l
12	EUS	subtype	baseline (s1) / boxmap	n/l	0.193	n/l
12	EUS	subtype	baseline (s2) / full	n/l	0.301	n/l
12	EUS	subtype	baseline (s2) / bg_patch	n/l	0.254	n/l
12	EUS	subtype	baseline (s2) / boxmap	n/l	0.200	n/l
12	EUS	subtype	baseline (s3) / full	n/l	0.336	n/l
12	EUS	subtype	baseline (s3) / bg_patch	n/l	0.292	n/l
12	EUS	subtype	baseline (s3) / boxmap	n/l	0.202	n/l
12	EUS	subtype	baseline (s4) / full	n/l	0.301	n/l
12	EUS	subtype	baseline (s4) / bg_patch	n/l	0.253	n/l
12	EUS	subtype	baseline (s4) / boxmap	n/l	0.200	n/l
12	WLE	binary	baseline_100ep (s42) / full	n/l	0.715	n/l
12	WLE	binary	baseline_100ep (s42) / bg_patch	n/l	0.663	n/l
12	WLE	binary	baseline_100ep (s42) / boxmap	n/l	0.500	n/l
12	WLE	binary	baseline_100ep (s7) / full	n/l	0.709	n/l
12	WLE	binary	baseline_100ep (s7) / bg_patch	n/l	0.580	n/l
12	WLE	binary	baseline_100ep (s7) / boxmap	n/l	0.500	n/l
12	WLE	binary	baseline_100ep (s123) / full	n/l	0.714	n/l
12	WLE	binary	baseline_100ep (s123) / bg_patch	n/l	0.628	n/l
12	WLE	binary	baseline_100ep (s123) / boxmap	n/l	0.513	n/l
12	WLE	binary	baseline_100ep (s1) / full	n/l	0.727	n/l
12	WLE	binary	baseline_100ep (s1) / bg_patch	n/l	0.629	n/l
12	WLE	binary	baseline_100ep (s1) / boxmap	n/l	0.500	n/l
12	WLE	binary	baseline_100ep (s2) / full	n/l	0.720	n/l
12	WLE	binary	baseline_100ep (s2) / bg_patch	n/l	0.610	n/l
12	WLE	binary	baseline_100ep (s2) / boxmap	n/l	0.488	n/l
12	WLE	binary	baseline_100ep (s3) / full	n/l	0.741	n/l
12	WLE	binary	baseline_100ep (s3) / bg_patch	n/l	0.640	n/l
12	WLE	binary	baseline_100ep (s3) / boxmap	n/l	0.463	n/l
12	WLE	binary	baseline_100ep (s4) / full	n/l	0.720	n/l
12	WLE	binary	baseline_100ep (s4) / bg_patch	n/l	0.608	n/l
12	WLE	binary	baseline_100ep (s4) / boxmap	n/l	0.543	n/l
12	WLE	subtype	baseline_100ep (s42) / full	n/l	0.349	n/l
12	WLE	subtype	baseline_100ep (s42) / bg_patch	n/l	0.222	n/l
12	WLE	subtype	baseline_100ep (s42) / boxmap	n/l	0.236	n/l
12	WLE	subtype	baseline_100ep (s7) / full	n/l	0.359	n/l
12	WLE	subtype	baseline_100ep (s7) / bg_patch	n/l	0.255	n/l
12	WLE	subtype	baseline_100ep (s7) / boxmap	n/l	0.200	n/l

Exp.	Mod.	Task	Configuration	Acc.	Bal. Acc.	P/R/F1/ROC/PR-AUC
12	WLE	subtype	baseline_100ep (s123) / full	n/l	0.397	n/l
12	WLE	subtype	baseline_100ep (s123) / bg_patch	n/l	0.291	n/l
12	WLE	subtype	baseline_100ep (s123) / boxmap	n/l	0.206	n/l
12	WLE	subtype	baseline_100ep (s1) / full	n/l	0.363	n/l
12	WLE	subtype	baseline_100ep (s1) / bg_patch	n/l	0.210	n/l
12	WLE	subtype	baseline_100ep (s1) / boxmap	n/l	0.201	n/l
12	WLE	subtype	baseline_100ep (s2) / full	n/l	0.402	n/l
12	WLE	subtype	baseline_100ep (s2) / bg_patch	n/l	0.291	n/l
12	WLE	subtype	baseline_100ep (s2) / boxmap	n/l	0.197	n/l
12	WLE	subtype	baseline_100ep (s3) / full	n/l	0.391	n/l
12	WLE	subtype	baseline_100ep (s3) / bg_patch	n/l	0.281	n/l
12	WLE	subtype	baseline_100ep (s3) / boxmap	n/l	0.162	n/l
12	WLE	subtype	baseline_100ep (s4) / full	n/l	0.384	n/l
12	WLE	subtype	baseline_100ep (s4) / bg_patch	n/l	0.268	n/l
12	WLE	subtype	baseline_100ep (s4) / boxmap	n/l	0.244	n/l
12	EUS	binary	baseline_100ep (s42) / full	n/l	0.624	n/l
12	EUS	binary	baseline_100ep (s42) / bg_patch	n/l	0.578	n/l
12	EUS	binary	baseline_100ep (s42) / boxmap	n/l	0.624	n/l
12	EUS	binary	baseline_100ep (s7) / full	n/l	0.630	n/l
12	EUS	binary	baseline_100ep (s7) / bg_patch	n/l	0.585	n/l
12	EUS	binary	baseline_100ep (s7) / boxmap	n/l	0.500	n/l
12	EUS	binary	baseline_100ep (s123) / full	n/l	0.747	n/l
12	EUS	binary	baseline_100ep (s123) / bg_patch	n/l	0.626	n/l
12	EUS	binary	baseline_100ep (s123) / boxmap	n/l	0.500	n/l
12	EUS	binary	baseline_100ep (s1) / full	n/l	0.701	n/l
12	EUS	binary	baseline_100ep (s1) / bg_patch	n/l	0.555	n/l
12	EUS	binary	baseline_100ep (s1) / boxmap	n/l	0.500	n/l
12	EUS	binary	baseline_100ep (s2) / full	n/l	0.668	n/l
12	EUS	binary	baseline_100ep (s2) / bg_patch	n/l	0.544	n/l
12	EUS	binary	baseline_100ep (s2) / boxmap	n/l	0.500	n/l
12	EUS	binary	baseline_100ep (s3) / full	n/l	0.632	n/l
12	EUS	binary	baseline_100ep (s3) / bg_patch	n/l	0.545	n/l
12	EUS	binary	baseline_100ep (s3) / boxmap	n/l	0.500	n/l
12	EUS	binary	baseline_100ep (s4) / full	n/l	0.727	n/l
12	EUS	binary	baseline_100ep (s4) / bg_patch	n/l	0.605	n/l
12	EUS	binary	baseline_100ep (s4) / boxmap	n/l	0.500	n/l
12	EUS	subtype	baseline_100ep (s42) / full	n/l	0.436	n/l
12	EUS	subtype	baseline_100ep (s42) / bg_patch	n/l	0.272	n/l
12	EUS	subtype	baseline_100ep (s42) / boxmap	n/l	0.200	n/l
12	EUS	subtype	baseline_100ep (s7) / full	n/l	0.297	n/l
12	EUS	subtype	baseline_100ep (s7) / bg_patch	n/l	0.263	n/l
12	EUS	subtype	baseline_100ep (s7) / boxmap	n/l	0.199	n/l
12	EUS	subtype	baseline_100ep (s123) / full	n/l	0.311	n/l
12	EUS	subtype	baseline_100ep (s123) / bg_patch	n/l	0.254	n/l
12	EUS	subtype	baseline_100ep (s123) / boxmap	n/l	0.199	n/l
12	EUS	subtype	baseline_100ep (s1) / full	n/l	0.388	n/l
12	EUS	subtype	baseline_100ep (s1) / bg_patch	n/l	0.283	n/l
12	EUS	subtype	baseline_100ep (s1) / boxmap	n/l	0.194	n/l
12	EUS	subtype	baseline_100ep (s2) / full	n/l	0.314	n/l
12	EUS	subtype	baseline_100ep (s2) / bg_patch	n/l	0.271	n/l
12	EUS	subtype	baseline_100ep (s2) / boxmap	n/l	0.158	n/l
12	EUS	subtype	baseline_100ep (s3) / full	n/l	0.328	n/l
12	EUS	subtype	baseline_100ep (s3) / bg_patch	n/l	0.212	n/l
12	EUS	subtype	baseline_100ep (s3) / boxmap	n/l	0.189	n/l
12	EUS	subtype	baseline_100ep (s4) / full	n/l	0.378	n/l
12	EUS	subtype	baseline_100ep (s4) / bg_patch	n/l	0.287	n/l
12	EUS	subtype	baseline_100ep (s4) / boxmap	n/l	0.166	n/l
13	WLE	binary	blur_light (s42) / full	n/l	0.713	n/l
13	WLE	binary	blur_light (s42) / bg_patch	n/l	0.631	n/l
13	WLE	binary	blur_light (s42) / boxmap	n/l	0.483	n/l
13	WLE	binary	blur_light (s7) / full	n/l	0.744	n/l
13	WLE	binary	blur_light (s7) / bg_patch	n/l	0.619	n/l
13	WLE	binary	blur_light (s7) / boxmap	n/l	0.502	n/l
13	WLE	binary	blur_light (s123) / full	n/l	0.732	n/l
13	WLE	binary	blur_light (s123) / bg_patch	n/l	0.642	n/l
13	WLE	binary	blur_light (s123) / boxmap	n/l	0.510	n/l
13	WLE	binary	blur_heavy (s42) / full	n/l	0.715	n/l
13	WLE	binary	blur_heavy (s42) / bg_patch	n/l	0.593	n/l

Exp.	Mod.	Task	Configuration	Acc.	Bal. Acc.	P/R/F1/ROC/PR-AUC
13	WLE	binary	blur_heavy (s42) / boxmap	n/l	0.493	n/l
13	WLE	binary	blur_heavy (s7) / full	n/l	0.716	n/l
13	WLE	binary	blur_heavy (s7) / bg_patch	n/l	0.625	n/l
13	WLE	binary	blur_heavy (s7) / boxmap	n/l	0.498	n/l
13	WLE	binary	blur_heavy (s123) / full	n/l	0.720	n/l
13	WLE	binary	blur_heavy (s123) / bg_patch	n/l	0.601	n/l
13	WLE	binary	blur_heavy (s123) / boxmap	n/l	0.506	n/l
13	WLE	subtype	blur_light (s42) / full	n/l	0.332	n/l
13	WLE	subtype	blur_light (s42) / bg_patch	n/l	0.286	n/l
13	WLE	subtype	blur_light (s42) / boxmap	n/l	0.174	n/l
13	WLE	subtype	blur_light (s7) / full	n/l	0.403	n/l
13	WLE	subtype	blur_light (s7) / bg_patch	n/l	0.285	n/l
13	WLE	subtype	blur_light (s7) / boxmap	n/l	0.245	n/l
13	WLE	subtype	blur_light (s123) / full	n/l	0.350	n/l
13	WLE	subtype	blur_light (s123) / bg_patch	n/l	0.266	n/l
13	WLE	subtype	blur_light (s123) / boxmap	n/l	0.200	n/l
13	WLE	subtype	blur_heavy (s42) / full	n/l	0.335	n/l
13	WLE	subtype	blur_heavy (s42) / bg_patch	n/l	0.259	n/l
13	WLE	subtype	blur_heavy (s42) / boxmap	n/l	0.239	n/l
13	WLE	subtype	blur_heavy (s7) / full	n/l	0.421	n/l
13	WLE	subtype	blur_heavy (s7) / bg_patch	n/l	0.292	n/l
13	WLE	subtype	blur_heavy (s7) / boxmap	n/l	0.220	n/l
13	WLE	subtype	blur_heavy (s123) / full	n/l	0.401	n/l
13	WLE	subtype	blur_heavy (s123) / bg_patch	n/l	0.246	n/l
13	WLE	subtype	blur_heavy (s123) / boxmap	n/l	0.200	n/l
13	EUS	binary	blur_light (s42) / full	n/l	0.638	n/l
13	EUS	binary	blur_light (s42) / bg_patch	n/l	0.568	n/l
13	EUS	binary	blur_light (s42) / boxmap	n/l	0.369	n/l
13	EUS	binary	blur_light (s7) / full	n/l	0.619	n/l
13	EUS	binary	blur_light (s7) / bg_patch	n/l	0.635	n/l
13	EUS	binary	blur_light (s7) / boxmap	n/l	0.645	n/l
13	EUS	binary	blur_light (s123) / full	n/l	0.627	n/l
13	EUS	binary	blur_light (s123) / bg_patch	n/l	0.580	n/l
13	EUS	binary	blur_light (s123) / boxmap	n/l	0.500	n/l
13	EUS	binary	blur_heavy (s42) / full	n/l	0.613	n/l
13	EUS	binary	blur_heavy (s42) / bg_patch	n/l	0.618	n/l
13	EUS	binary	blur_heavy (s42) / boxmap	n/l	0.486	n/l
13	EUS	binary	blur_heavy (s7) / full	n/l	0.586	n/l
13	EUS	binary	blur_heavy (s7) / bg_patch	n/l	0.543	n/l
13	EUS	binary	blur_heavy (s7) / boxmap	n/l	0.500	n/l
13	EUS	binary	blur_heavy (s123) / full	n/l	0.667	n/l
13	EUS	binary	blur_heavy (s123) / bg_patch	n/l	0.635	n/l
13	EUS	binary	blur_heavy (s123) / boxmap	n/l	0.405	n/l
13	EUS	subtype	blur_light (s42) / full	n/l	0.349	n/l
13	EUS	subtype	blur_light (s42) / bg_patch	n/l	0.238	n/l
13	EUS	subtype	blur_light (s42) / boxmap	n/l	0.190	n/l
13	EUS	subtype	blur_light (s7) / full	n/l	0.293	n/l
13	EUS	subtype	blur_light (s7) / bg_patch	n/l	0.249	n/l
13	EUS	subtype	blur_light (s7) / boxmap	n/l	0.181	n/l
13	EUS	subtype	blur_light (s123) / full	n/l	0.315	n/l
13	EUS	subtype	blur_light (s123) / bg_patch	n/l	0.258	n/l
13	EUS	subtype	blur_light (s123) / boxmap	n/l	0.202	n/l
13	EUS	subtype	blur_heavy (s42) / full	n/l	0.332	n/l
13	EUS	subtype	blur_heavy (s42) / bg_patch	n/l	0.316	n/l
13	EUS	subtype	blur_heavy (s42) / boxmap	n/l	0.139	n/l
13	EUS	subtype	blur_heavy (s7) / full	n/l	0.336	n/l
13	EUS	subtype	blur_heavy (s7) / bg_patch	n/l	0.264	n/l
13	EUS	subtype	blur_heavy (s7) / boxmap	n/l	0.193	n/l
13	EUS	subtype	blur_heavy (s123) / full	n/l	0.353	n/l
13	EUS	subtype	blur_heavy (s123) / bg_patch	n/l	0.327	n/l
13	EUS	subtype	blur_heavy (s123) / boxmap	n/l	0.200	n/l
14	WLE	binary	se_layer4_ce (s42) / full	n/l	0.710	n/l
14	WLE	binary	se_layer4_ce (s42) / bg_patch	n/l	0.634	n/l
14	WLE	binary	se_layer4_ce (s42) / boxmap	n/l	0.490	n/l
14	WLE	binary	se_layer4_ce (s7) / full	n/l	0.721	n/l
14	WLE	binary	se_layer4_ce (s7) / bg_patch	n/l	0.653	n/l
14	WLE	binary	se_layer4_ce (s7) / boxmap	n/l	0.477	n/l
14	WLE	binary	se_layer4_ce (s123) / full	n/l	0.730	n/l

Exp.	Mod.	Task	Configuration	Acc.	Bal. Acc.	P/R/F1/ROC/PR-AUC
14	WLE	binary	se_layer4_ce (s123) / bg_patch	n/l	0.602	n/l
14	WLE	binary	se_layer4_ce (s123) / boxmap	n/l	0.480	n/l
14	WLE	binary	se_all_ce (s42) / full	n/l	0.717	n/l
14	WLE	binary	se_all_ce (s42) / bg_patch	n/l	0.653	n/l
14	WLE	binary	se_all_ce (s42) / boxmap	n/l	0.506	n/l
14	WLE	binary	se_all_ce (s7) / full	n/l	0.711	n/l
14	WLE	binary	se_all_ce (s7) / bg_patch	n/l	0.617	n/l
14	WLE	binary	se_all_ce (s7) / boxmap	n/l	0.493	n/l
14	WLE	binary	se_all_ce (s123) / full	n/l	0.717	n/l
14	WLE	binary	se_all_ce (s123) / bg_patch	n/l	0.615	n/l
14	WLE	binary	se_all_ce (s123) / boxmap	n/l	0.500	n/l
14	WLE	binary	se_layer4_adv (s42) / full	n/l	0.703	n/l
14	WLE	binary	se_layer4_adv (s42) / bg_patch	n/l	0.581	n/l
14	WLE	binary	se_layer4_adv (s42) / boxmap	n/l	0.498	n/l
14	WLE	binary	se_layer4_adv (s7) / full	n/l	0.705	n/l
14	WLE	binary	se_layer4_adv (s7) / bg_patch	n/l	0.595	n/l
14	WLE	binary	se_layer4_adv (s7) / boxmap	n/l	0.479	n/l
14	WLE	binary	se_layer4_adv (s123) / full	n/l	0.696	n/l
14	WLE	binary	se_layer4_adv (s123) / bg_patch	n/l	0.613	n/l
14	WLE	binary	se_layer4_adv (s123) / boxmap	n/l	0.467	n/l
14	WLE	binary	se_all_adv (s42) / full	n/l	0.688	n/l
14	WLE	binary	se_all_adv (s42) / bg_patch	n/l	0.539	n/l
14	WLE	binary	se_all_adv (s42) / boxmap	n/l	0.494	n/l
14	WLE	binary	se_all_adv (s7) / full	n/l	0.689	n/l
14	WLE	binary	se_all_adv (s7) / bg_patch	n/l	0.601	n/l
14	WLE	binary	se_all_adv (s7) / boxmap	n/l	0.557	n/l
14	WLE	binary	se_all_adv (s123) / full	n/l	0.702	n/l
14	WLE	binary	se_all_adv (s123) / bg_patch	n/l	0.593	n/l
14	WLE	binary	se_all_adv (s123) / boxmap	n/l	0.500	n/l
14	WLE	subtype	se_layer4_ce (s42) / full	n/l	0.354	n/l
14	WLE	subtype	se_layer4_ce (s42) / bg_patch	n/l	0.281	n/l
14	WLE	subtype	se_layer4_ce (s42) / boxmap	n/l	0.208	n/l
14	WLE	subtype	se_layer4_ce (s7) / full	n/l	0.378	n/l
14	WLE	subtype	se_layer4_ce (s7) / bg_patch	n/l	0.289	n/l
14	WLE	subtype	se_layer4_ce (s7) / boxmap	n/l	0.223	n/l
14	WLE	subtype	se_layer4_ce (s123) / full	n/l	0.388	n/l
14	WLE	subtype	se_layer4_ce (s123) / bg_patch	n/l	0.256	n/l
14	WLE	subtype	se_layer4_ce (s123) / boxmap	n/l	0.193	n/l
14	WLE	subtype	se_all_ce (s42) / full	n/l	0.341	n/l
14	WLE	subtype	se_all_ce (s42) / bg_patch	n/l	0.295	n/l
14	WLE	subtype	se_all_ce (s42) / boxmap	n/l	0.204	n/l
14	WLE	subtype	se_all_ce (s7) / full	n/l	0.452	n/l
14	WLE	subtype	se_all_ce (s7) / bg_patch	n/l	0.305	n/l
14	WLE	subtype	se_all_ce (s7) / boxmap	n/l	0.189	n/l
14	WLE	subtype	se_all_ce (s123) / full	n/l	0.377	n/l
14	WLE	subtype	se_all_ce (s123) / bg_patch	n/l	0.257	n/l
14	WLE	subtype	se_all_ce (s123) / boxmap	n/l	0.237	n/l
14	WLE	subtype	se_layer4_adv (s42) / full	n/l	0.364	n/l
14	WLE	subtype	se_layer4_adv (s42) / bg_patch	n/l	0.278	n/l
14	WLE	subtype	se_layer4_adv (s42) / boxmap	n/l	0.201	n/l
14	WLE	subtype	se_layer4_adv (s7) / full	n/l	0.411	n/l
14	WLE	subtype	se_layer4_adv (s7) / bg_patch	n/l	0.271	n/l
14	WLE	subtype	se_layer4_adv (s7) / boxmap	n/l	0.193	n/l
14	WLE	subtype	se_layer4_adv (s123) / full	n/l	0.364	n/l
14	WLE	subtype	se_layer4_adv (s123) / bg_patch	n/l	0.227	n/l
14	WLE	subtype	se_layer4_adv (s123) / boxmap	n/l	0.202	n/l
14	WLE	subtype	se_all_adv (s42) / full	n/l	0.357	n/l
14	WLE	subtype	se_all_adv (s42) / bg_patch	n/l	0.219	n/l
14	WLE	subtype	se_all_adv (s42) / boxmap	n/l	0.200	n/l
14	WLE	subtype	se_all_adv (s7) / full	n/l	0.413	n/l
14	WLE	subtype	se_all_adv (s7) / bg_patch	n/l	0.299	n/l
14	WLE	subtype	se_all_adv (s7) / boxmap	n/l	0.172	n/l
14	WLE	subtype	se_all_adv (s123) / full	n/l	0.370	n/l
14	WLE	subtype	se_all_adv (s123) / bg_patch	n/l	0.239	n/l
14	WLE	subtype	se_all_adv (s123) / boxmap	n/l	0.257	n/l
14	EUS	binary	se_layer4_ce (s42) / full	n/l	0.627	n/l
14	EUS	binary	se_layer4_ce (s42) / bg_patch	n/l	0.485	n/l
14	EUS	binary	se_layer4_ce (s42) / boxmap	n/l	0.500	n/l

Exp.	Mod.	Task	Configuration	Acc.	Bal. Acc.	P/R/F1/ROC/PR-AUC
14	EUS	binary	se_layer4_ce (s7) / full	n/l	0.649	n/l
14	EUS	binary	se_layer4_ce (s7) / bg_patch	n/l	0.537	n/l
14	EUS	binary	se_layer4_ce (s7) / boxmap	n/l	0.500	n/l
14	EUS	binary	se_layer4_ce (s123) / full	n/l	0.620	n/l
14	EUS	binary	se_layer4_ce (s123) / bg_patch	n/l	0.513	n/l
14	EUS	binary	se_layer4_ce (s123) / boxmap	n/l	0.500	n/l
14	EUS	binary	se_all_ce (s42) / full	n/l	0.592	n/l
14	EUS	binary	se_all_ce (s42) / bg_patch	n/l	0.619	n/l
14	EUS	binary	se_all_ce (s42) / boxmap	n/l	0.455	n/l
14	EUS	binary	se_all_ce (s7) / full	n/l	0.662	n/l
14	EUS	binary	se_all_ce (s7) / bg_patch	n/l	0.556	n/l
14	EUS	binary	se_all_ce (s7) / boxmap	n/l	0.500	n/l
14	EUS	binary	se_all_ce (s123) / full	n/l	0.615	n/l
14	EUS	binary	se_all_ce (s123) / bg_patch	n/l	0.606	n/l
14	EUS	binary	se_all_ce (s123) / boxmap	n/l	0.500	n/l
14	EUS	binary	se_layer4_adv (s42) / full	n/l	0.640	n/l
14	EUS	binary	se_layer4_adv (s42) / bg_patch	n/l	0.511	n/l
14	EUS	binary	se_layer4_adv (s42) / boxmap	n/l	0.500	n/l
14	EUS	binary	se_layer4_adv (s7) / full	n/l	0.639	n/l
14	EUS	binary	se_layer4_adv (s7) / bg_patch	n/l	0.475	n/l
14	EUS	binary	se_layer4_adv (s7) / boxmap	n/l	0.500	n/l
14	EUS	binary	se_layer4_adv (s123) / full	n/l	0.613	n/l
14	EUS	binary	se_layer4_adv (s123) / bg_patch	n/l	0.629	n/l
14	EUS	binary	se_layer4_adv (s123) / boxmap	n/l	0.500	n/l
14	EUS	binary	se_all_adv (s42) / full	n/l	0.608	n/l
14	EUS	binary	se_all_adv (s42) / bg_patch	n/l	0.496	n/l
14	EUS	binary	se_all_adv (s42) / boxmap	n/l	0.500	n/l
14	EUS	binary	se_all_adv (s7) / full	n/l	0.620	n/l
14	EUS	binary	se_all_adv (s7) / bg_patch	n/l	0.557	n/l
14	EUS	binary	se_all_adv (s7) / boxmap	n/l	0.496	n/l
14	EUS	binary	se_all_adv (s123) / full	n/l	0.638	n/l
14	EUS	binary	se_all_adv (s123) / bg_patch	n/l	0.515	n/l
14	EUS	binary	se_all_adv (s123) / boxmap	n/l	0.500	n/l
14	EUS	subtype	se_layer4_ce (s42) / full	n/l	0.357	n/l
14	EUS	subtype	se_layer4_ce (s42) / bg_patch	n/l	0.273	n/l
14	EUS	subtype	se_layer4_ce (s42) / boxmap	n/l	0.200	n/l
14	EUS	subtype	se_layer4_ce (s7) / full	n/l	0.277	n/l
14	EUS	subtype	se_layer4_ce (s7) / bg_patch	n/l	0.250	n/l
14	EUS	subtype	se_layer4_ce (s7) / boxmap	n/l	0.228	n/l
14	EUS	subtype	se_layer4_ce (s123) / full	n/l	0.379	n/l
14	EUS	subtype	se_layer4_ce (s123) / bg_patch	n/l	0.231	n/l
14	EUS	subtype	se_layer4_ce (s123) / boxmap	n/l	0.200	n/l
14	EUS	subtype	se_all_ce (s42) / full	n/l	0.274	n/l
14	EUS	subtype	se_all_ce (s42) / bg_patch	n/l	0.256	n/l
14	EUS	subtype	se_all_ce (s42) / boxmap	n/l	0.202	n/l
14	EUS	subtype	se_all_ce (s7) / full	n/l	0.292	n/l
14	EUS	subtype	se_all_ce (s7) / bg_patch	n/l	0.231	n/l
14	EUS	subtype	se_all_ce (s7) / boxmap	n/l	0.194	n/l
14	EUS	subtype	se_all_ce (s123) / full	n/l	0.376	n/l
14	EUS	subtype	se_all_ce (s123) / bg_patch	n/l	0.316	n/l
14	EUS	subtype	se_all_ce (s123) / boxmap	n/l	0.200	n/l
14	EUS	subtype	se_layer4_adv (s42) / full	n/l	0.332	n/l
14	EUS	subtype	se_layer4_adv (s42) / bg_patch	n/l	0.250	n/l
14	EUS	subtype	se_layer4_adv (s42) / boxmap	n/l	0.182	n/l
14	EUS	subtype	se_layer4_adv (s7) / full	n/l	0.265	n/l
14	EUS	subtype	se_layer4_adv (s7) / bg_patch	n/l	0.247	n/l
14	EUS	subtype	se_layer4_adv (s7) / boxmap	n/l	0.200	n/l
14	EUS	subtype	se_layer4_adv (s123) / full	n/l	0.363	n/l
14	EUS	subtype	se_layer4_adv (s123) / bg_patch	n/l	0.252	n/l
14	EUS	subtype	se_layer4_adv (s123) / boxmap	n/l	0.200	n/l
14	EUS	subtype	se_all_adv (s42) / full	n/l	0.316	n/l
14	EUS	subtype	se_all_adv (s42) / bg_patch	n/l	0.233	n/l
14	EUS	subtype	se_all_adv (s42) / boxmap	n/l	0.200	n/l
14	EUS	subtype	se_all_adv (s7) / full	n/l	0.333	n/l
14	EUS	subtype	se_all_adv (s7) / bg_patch	n/l	0.189	n/l
14	EUS	subtype	se_all_adv (s7) / boxmap	n/l	0.206	n/l
14	EUS	subtype	se_all_adv (s123) / full	n/l	0.355	n/l
14	EUS	subtype	se_all_adv (s123) / bg_patch	n/l	0.287	n/l

Exp.	Mod.	Task	Configuration	Acc.	Bal. Acc.	P/R/F1/ROC/PR-AUC
14	EUS	subtype	se_all_adv (s123) / boxmap	n/l	0.200	n/l
14	WLE	binary	se_all_adv (s1) / full	n/l	0.704	n/l
14	WLE	binary	se_all_adv (s1) / bg_patch	n/l	0.558	n/l
14	WLE	binary	se_all_adv (s1) / boxmap	n/l	0.450	n/l
14	WLE	binary	se_all_adv (s2) / full	n/l	0.710	n/l
14	WLE	binary	se_all_adv (s2) / bg_patch	n/l	0.583	n/l
14	WLE	binary	se_all_adv (s2) / boxmap	n/l	0.508	n/l
14	WLE	binary	se_all_adv (s3) / full	n/l	0.680	n/l
14	WLE	binary	se_all_adv (s3) / bg_patch	n/l	0.567	n/l
14	WLE	binary	se_all_adv (s3) / boxmap	n/l	0.488	n/l
14	WLE	binary	se_all_adv (s4) / full	n/l	0.709	n/l
14	WLE	binary	se_all_adv (s4) / bg_patch	n/l	0.548	n/l
14	WLE	binary	se_all_adv (s4) / boxmap	n/l	0.497	n/l
14	WLE	subtype	se_all_adv (s1) / full	n/l	0.363	n/l
14	WLE	subtype	se_all_adv (s1) / bg_patch	n/l	0.249	n/l
14	WLE	subtype	se_all_adv (s1) / boxmap	n/l	0.197	n/l
14	WLE	subtype	se_all_adv (s2) / full	n/l	0.344	n/l
14	WLE	subtype	se_all_adv (s2) / bg_patch	n/l	0.229	n/l
14	WLE	subtype	se_all_adv (s2) / boxmap	n/l	0.166	n/l
14	WLE	subtype	se_all_adv (s3) / full	n/l	0.444	n/l
14	WLE	subtype	se_all_adv (s3) / bg_patch	n/l	0.274	n/l
14	WLE	subtype	se_all_adv (s3) / boxmap	n/l	0.202	n/l
14	WLE	subtype	se_all_adv (s4) / full	n/l	0.401	n/l
14	WLE	subtype	se_all_adv (s4) / bg_patch	n/l	0.276	n/l
14	WLE	subtype	se_all_adv (s4) / boxmap	n/l	0.166	n/l
14	EUS	binary	se_all_adv (s1) / full	n/l	0.634	n/l
14	EUS	binary	se_all_adv (s1) / bg_patch	n/l	0.482	n/l
14	EUS	binary	se_all_adv (s1) / boxmap	n/l	0.500	n/l
14	EUS	binary	se_all_adv (s2) / full	n/l	0.678	n/l
14	EUS	binary	se_all_adv (s2) / bg_patch	n/l	0.599	n/l
14	EUS	binary	se_all_adv (s2) / boxmap	n/l	0.456	n/l
14	EUS	binary	se_all_adv (s3) / full	n/l	0.622	n/l
14	EUS	binary	se_all_adv (s3) / bg_patch	n/l	0.532	n/l
14	EUS	binary	se_all_adv (s3) / boxmap	n/l	0.500	n/l
14	EUS	binary	se_all_adv (s4) / full	n/l	0.606	n/l
14	EUS	binary	se_all_adv (s4) / bg_patch	n/l	0.485	n/l
14	EUS	binary	se_all_adv (s4) / boxmap	n/l	0.500	n/l
14	EUS	subtype	se_all_adv (s1) / full	n/l	0.345	n/l
14	EUS	subtype	se_all_adv (s1) / bg_patch	n/l	0.312	n/l
14	EUS	subtype	se_all_adv (s1) / boxmap	n/l	0.199	n/l
14	EUS	subtype	se_all_adv (s2) / full	n/l	0.336	n/l
14	EUS	subtype	se_all_adv (s2) / bg_patch	n/l	0.271	n/l
14	EUS	subtype	se_all_adv (s2) / boxmap	n/l	0.212	n/l
14	EUS	subtype	se_all_adv (s3) / full	n/l	0.281	n/l
14	EUS	subtype	se_all_adv (s3) / bg_patch	n/l	0.277	n/l
14	EUS	subtype	se_all_adv (s3) / boxmap	n/l	0.310	n/l
14	EUS	subtype	se_all_adv (s4) / full	n/l	0.317	n/l
14	EUS	subtype	se_all_adv (s4) / bg_patch	n/l	0.169	n/l
14	EUS	subtype	se_all_adv (s4) / boxmap	n/l	0.195	n/l
14	WLE	binary	se_all_adv_100ep (s42) / full	n/l	0.703	n/l
14	WLE	binary	se_all_adv_100ep (s42) / bg_patch	n/l	0.577	n/l
14	WLE	binary	se_all_adv_100ep (s42) / boxmap	n/l	0.456	n/l
14	WLE	binary	se_all_adv_100ep (s7) / full	n/l	0.713	n/l
14	WLE	binary	se_all_adv_100ep (s7) / bg_patch	n/l	0.550	n/l
14	WLE	binary	se_all_adv_100ep (s7) / boxmap	n/l	0.486	n/l
14	WLE	binary	se_all_adv_100ep (s123) / full	n/l	0.715	n/l
14	WLE	binary	se_all_adv_100ep (s123) / bg_patch	n/l	0.619	n/l
14	WLE	binary	se_all_adv_100ep (s123) / boxmap	n/l	0.500	n/l
14	WLE	binary	se_all_adv_100ep (s1) / full	n/l	0.708	n/l
14	WLE	binary	se_all_adv_100ep (s1) / bg_patch	n/l	0.589	n/l
14	WLE	binary	se_all_adv_100ep (s1) / boxmap	n/l	0.443	n/l
14	WLE	binary	se_all_adv_100ep (s2) / full	n/l	0.700	n/l
14	WLE	binary	se_all_adv_100ep (s2) / bg_patch	n/l	0.596	n/l
14	WLE	binary	se_all_adv_100ep (s2) / boxmap	n/l	0.472	n/l
14	WLE	binary	se_all_adv_100ep (s3) / full	n/l	0.710	n/l
14	WLE	binary	se_all_adv_100ep (s3) / bg_patch	n/l	0.590	n/l
14	WLE	binary	se_all_adv_100ep (s3) / boxmap	n/l	0.500	n/l
14	WLE	binary	se_all_adv_100ep (s4) / full	n/l	0.724	n/l

Exp.	Mod.	Task	Configuration	Acc.	Bal. Acc.	P/R/F1/ROC/PR-AUC
14	WLE	binary	se_all_adv_100ep (s4) / bg_patch	n/l	0.551	n/l
14	WLE	binary	se_all_adv_100ep (s4) / boxmap	n/l	0.500	n/l
14	WLE	subtype	se_all_adv_100ep (s42) / full	n/l	0.363	n/l
14	WLE	subtype	se_all_adv_100ep (s42) / bg_patch	n/l	0.228	n/l
14	WLE	subtype	se_all_adv_100ep (s42) / boxmap	n/l	0.212	n/l
14	WLE	subtype	se_all_adv_100ep (s7) / full	n/l	0.386	n/l
14	WLE	subtype	se_all_adv_100ep (s7) / bg_patch	n/l	0.232	n/l
14	WLE	subtype	se_all_adv_100ep (s7) / boxmap	n/l	0.207	n/l
14	WLE	subtype	se_all_adv_100ep (s123) / full	n/l	0.367	n/l
14	WLE	subtype	se_all_adv_100ep (s123) / bg_patch	n/l	0.250	n/l
14	WLE	subtype	se_all_adv_100ep (s123) / boxmap	n/l	0.258	n/l
14	WLE	subtype	se_all_adv_100ep (s1) / full	n/l	0.362	n/l
14	WLE	subtype	se_all_adv_100ep (s1) / bg_patch	n/l	0.299	n/l
14	WLE	subtype	se_all_adv_100ep (s1) / boxmap	n/l	0.201	n/l
14	WLE	subtype	se_all_adv_100ep (s2) / full	n/l	0.406	n/l
14	WLE	subtype	se_all_adv_100ep (s2) / bg_patch	n/l	0.270	n/l
14	WLE	subtype	se_all_adv_100ep (s2) / boxmap	n/l	0.244	n/l
14	WLE	subtype	se_all_adv_100ep (s3) / full	n/l	0.383	n/l
14	WLE	subtype	se_all_adv_100ep (s3) / bg_patch	n/l	0.203	n/l
14	WLE	subtype	se_all_adv_100ep (s3) / boxmap	n/l	0.194	n/l
14	WLE	subtype	se_all_adv_100ep (s4) / full	n/l	0.387	n/l
14	WLE	subtype	se_all_adv_100ep (s4) / bg_patch	n/l	0.272	n/l
14	WLE	subtype	se_all_adv_100ep (s4) / boxmap	n/l	0.198	n/l
14	EUS	binary	se_all_adv_100ep (s42) / full	n/l	0.693	n/l
14	EUS	binary	se_all_adv_100ep (s42) / bg_patch	n/l	0.511	n/l
14	EUS	binary	se_all_adv_100ep (s42) / boxmap	n/l	0.499	n/l
14	EUS	binary	se_all_adv_100ep (s7) / full	n/l	0.631	n/l
14	EUS	binary	se_all_adv_100ep (s7) / bg_patch	n/l	0.538	n/l
14	EUS	binary	se_all_adv_100ep (s7) / boxmap	n/l	0.499	n/l
14	EUS	binary	se_all_adv_100ep (s123) / full	n/l	0.634	n/l
14	EUS	binary	se_all_adv_100ep (s123) / bg_patch	n/l	0.610	n/l
14	EUS	binary	se_all_adv_100ep (s123) / boxmap	n/l	0.500	n/l
14	EUS	binary	se_all_adv_100ep (s1) / full	n/l	0.661	n/l
14	EUS	binary	se_all_adv_100ep (s1) / bg_patch	n/l	0.516	n/l
14	EUS	binary	se_all_adv_100ep (s1) / boxmap	n/l	0.500	n/l
14	EUS	binary	se_all_adv_100ep (s2) / full	n/l	0.679	n/l
14	EUS	binary	se_all_adv_100ep (s2) / bg_patch	n/l	0.488	n/l
14	EUS	binary	se_all_adv_100ep (s2) / boxmap	n/l	0.493	n/l
14	EUS	binary	se_all_adv_100ep (s3) / full	n/l	0.646	n/l
14	EUS	binary	se_all_adv_100ep (s3) / bg_patch	n/l	0.553	n/l
14	EUS	binary	se_all_adv_100ep (s3) / boxmap	n/l	0.423	n/l
14	EUS	binary	se_all_adv_100ep (s4) / full	n/l	0.634	n/l
14	EUS	binary	se_all_adv_100ep (s4) / bg_patch	n/l	0.514	n/l
14	EUS	binary	se_all_adv_100ep (s4) / boxmap	n/l	0.500	n/l
14	EUS	subtype	se_all_adv_100ep (s42) / full	n/l	0.444	n/l
14	EUS	subtype	se_all_adv_100ep (s42) / bg_patch	n/l	0.226	n/l
14	EUS	subtype	se_all_adv_100ep (s42) / boxmap	n/l	0.200	n/l
14	EUS	subtype	se_all_adv_100ep (s7) / full	n/l	0.318	n/l
14	EUS	subtype	se_all_adv_100ep (s7) / bg_patch	n/l	0.271	n/l
14	EUS	subtype	se_all_adv_100ep (s7) / boxmap	n/l	0.200	n/l
14	EUS	subtype	se_all_adv_100ep (s123) / full	n/l	0.354	n/l
14	EUS	subtype	se_all_adv_100ep (s123) / bg_patch	n/l	0.304	n/l
14	EUS	subtype	se_all_adv_100ep (s123) / boxmap	n/l	0.200	n/l
14	EUS	subtype	se_all_adv_100ep (s1) / full	n/l	0.360	n/l
14	EUS	subtype	se_all_adv_100ep (s1) / bg_patch	n/l	0.191	n/l
14	EUS	subtype	se_all_adv_100ep (s1) / boxmap	n/l	0.200	n/l
14	EUS	subtype	se_all_adv_100ep (s2) / full	n/l	0.388	n/l
14	EUS	subtype	se_all_adv_100ep (s2) / bg_patch	n/l	0.307	n/l
14	EUS	subtype	se_all_adv_100ep (s2) / boxmap	n/l	0.201	n/l
14	EUS	subtype	se_all_adv_100ep (s3) / full	n/l	0.309	n/l
14	EUS	subtype	se_all_adv_100ep (s3) / bg_patch	n/l	0.220	n/l
14	EUS	subtype	se_all_adv_100ep (s3) / boxmap	n/l	0.200	n/l
14	EUS	subtype	se_all_adv_100ep (s4) / full	n/l	0.367	n/l
14	EUS	subtype	se_all_adv_100ep (s4) / bg_patch	n/l	0.255	n/l
14	EUS	subtype	se_all_adv_100ep (s4) / boxmap	n/l	0.169	n/l
17	WLE	binary	resnet / full	0.734	0.647	n/l
17	WLE	binary	resnet / lesion	0.727	0.657	n/l
17	WLE	binary	resnet / bg_mask	0.725	0.638	n/l

Exp.	Mod.	Task	Configuration	Acc.	Bal. Acc.	P/R/F1/ROC/PR-AUC
17	WLE	binary	resnet / bg_patch	0.706	0.633	n/l
17	WLE	binary	resnet / boxmap	0.648	0.597	n/l
17	WLE	binary	dinov3 / full	0.788	0.737	n/l
17	WLE	binary	dinov3 / lesion	0.780	0.731	n/l
17	WLE	binary	dinov3 / bg_mask	0.765	0.709	n/l
17	WLE	binary	dinov3 / bg_patch	0.736	0.679	n/l
17	WLE	binary	dinov3 / boxmap	0.664	0.619	n/l
17	WLE	subtype	resnet / full	0.607	0.298	n/l
17	WLE	subtype	resnet / lesion	0.597	0.347	n/l
17	WLE	subtype	resnet / bg_mask	0.618	0.314	n/l
17	WLE	subtype	resnet / bg_patch	0.576	0.293	n/l
17	WLE	subtype	resnet / boxmap	0.419	0.285	n/l
17	WLE	subtype	dinov3 / full	0.644	0.382	n/l
17	WLE	subtype	dinov3 / lesion	0.622	0.414	n/l
17	WLE	subtype	dinov3 / bg_mask	0.606	0.337	n/l
17	WLE	subtype	dinov3 / bg_patch	0.581	0.326	n/l
17	WLE	subtype	dinov3 / boxmap	0.339	0.333	n/l
17	EUS	binary	resnet / full	0.744	0.573	n/l
17	EUS	binary	resnet / lesion	0.778	0.610	n/l
17	EUS	binary	resnet / bg_mask	0.753	0.597	n/l
17	EUS	binary	resnet / bg_patch	0.728	0.539	n/l
17	EUS	binary	resnet / boxmap	0.731	0.608	n/l
17	EUS	binary	dinov3 / full	0.741	0.604	n/l
17	EUS	binary	dinov3 / lesion	0.787	0.666	n/l
17	EUS	binary	dinov3 / bg_mask	0.761	0.627	n/l
17	EUS	binary	dinov3 / bg_patch	0.723	0.589	n/l
17	EUS	binary	dinov3 / boxmap	0.701	0.623	n/l
17	EUS	subtype	resnet / full	0.644	0.239	n/l
17	EUS	subtype	resnet / lesion	0.627	0.308	n/l
17	EUS	subtype	resnet / bg_mask	0.638	0.267	n/l
17	EUS	subtype	resnet / bg_patch	0.624	0.234	n/l
17	EUS	subtype	resnet / boxmap	0.526	0.292	n/l
17	EUS	subtype	dinov3 / full	0.631	0.246	n/l
17	EUS	subtype	dinov3 / lesion	0.638	0.346	n/l
17	EUS	subtype	dinov3 / bg_mask	0.655	0.276	n/l
17	EUS	subtype	dinov3 / bg_patch	0.618	0.248	n/l
17	EUS	subtype	dinov3 / boxmap	0.433	0.321	n/l
18	WLE	binary	['Center-1'] / internal	0.778	0.722	n/l
18	WLE	binary	['Center-1'] / external	0.718	0.590	n/l
18	WLE	subtype	['Center-1'] / internal	0.544	0.321	n/l
18	WLE	subtype	['Center-1'] / external	0.592	0.281	n/l
18	WLE	binary	['Center-2'] / internal	0.783	0.719	n/l
18	WLE	binary	['Center-2'] / external	0.532	0.661	n/l
18	WLE	subtype	['Center-2'] / internal	0.561	0.375	n/l
18	WLE	subtype	['Center-2'] / external	0.500	0.401	n/l
18	WLE	binary	['Center-3'] / internal	0.790	0.764	n/l
18	WLE	binary	['Center-3'] / external	0.701	0.630	n/l
18	WLE	subtype	['Center-3'] / internal	0.358	0.359	n/l
18	WLE	subtype	['Center-3'] / external	0.674	0.342	n/l
18	WLE	binary	['Center-4', 'Center-5', 'Center-6'] / internal	0.796	0.741	n/l
18	WLE	binary	['Center-4', 'Center-5', 'Center-6'] / external	0.688	0.681	n/l
18	WLE	subtype	['Center-4', 'Center-5', 'Center-6'] / internal	0.628	0.365	n/l
18	WLE	subtype	['Center-4', 'Center-5', 'Center-6'] / external	0.542	0.335	n/l
18	EUS	binary	['Center-1'] / internal	0.776	0.698	n/l
18	EUS	binary	['Center-1'] / external	0.713	0.614	n/l
18	EUS	subtype	['Center-1'] / internal	0.448	0.298	n/l
18	EUS	subtype	['Center-1'] / external	0.572	0.294	n/l
18	EUS	binary	['Center-2'] / internal	0.725	0.669	n/l
18	EUS	binary	['Center-2'] / external	0.676	0.752	n/l
18	EUS	subtype	['Center-2'] / internal	0.513	0.288	n/l
18	EUS	subtype	['Center-2'] / external	0.515	0.421	n/l
18	EUS	binary	['Center-3'] / internal	0.772	0.689	n/l
18	EUS	binary	['Center-3'] / external	0.759	0.690	n/l
18	EUS	subtype	['Center-3'] / internal	0.558	0.322	n/l

Exp.	Mod.	Task	Configuration	Acc.	Bal. Acc.	P/R/F1/ROC/PR-AUC
18	EUS	subtype	['Center-3'] / external	0.575	0.352	n/l